Roma Tre

Information Visualization

Multivariate Data Representation

M. Patrignani

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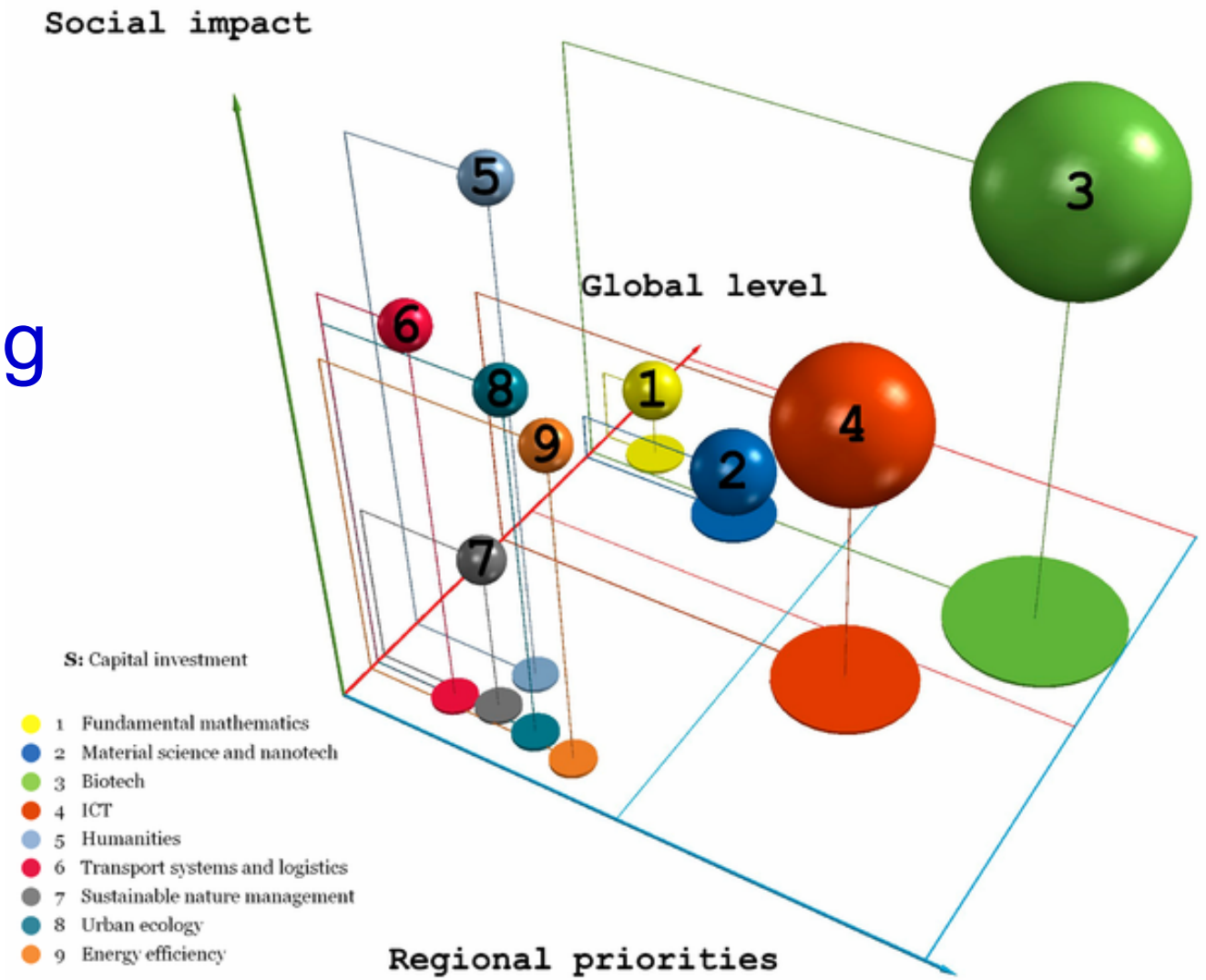
Overview

- Representation strategies for multivariate data
- Combined views
  - multiple bars, stacked bars, multiple views, table lenses, scatterplot matrix
- Other paradigms
  - multidimensional icons
  - alternative coordinate systems

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What about multivariate data?

- When the number of variables is more than three visualization is much more challenging
  - an example 3D bubble chart
    - variables: four quantitative plus one categorical



http://commons.wikimedia.org/wiki/File:3D_Bubble_Chart_Example.png

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Multivariate visualization strategies

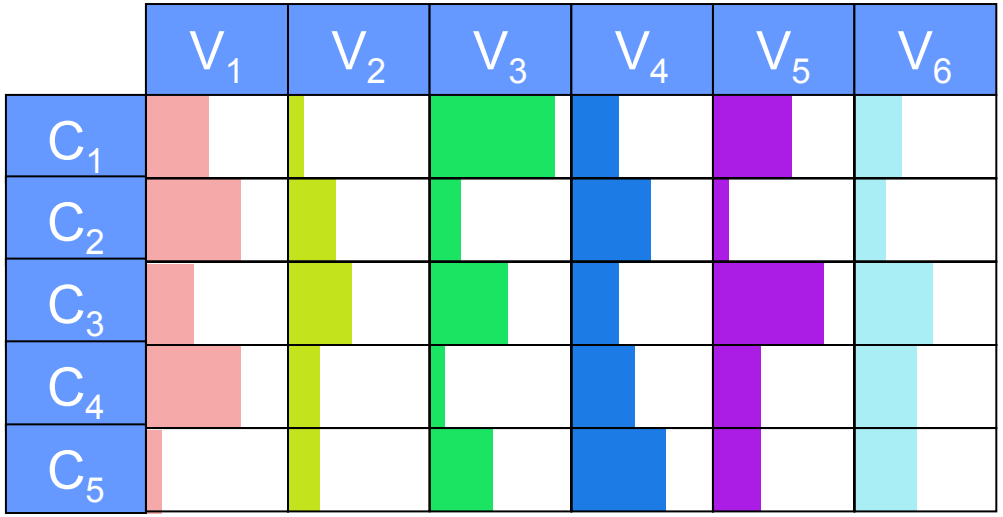
- Matrix based
  - table lens
- Bar charts variations
  - multiple bars, stacked bars, spineplots, mosaic plots
- Replication based
  - multiple view, scatterplot matrix
- Icons or glyphs based
  - Chernoff faces, multidimensional icons
- Alternative coordinate systems
  - parallel coordinates, star plots, star coordinates

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Table lens

- Basic idea [Rao & Card, 94]
  - start from a spreadsheet visualization
  - make the data more visual and symbolic
  - leverage basic bar chart effectiveness

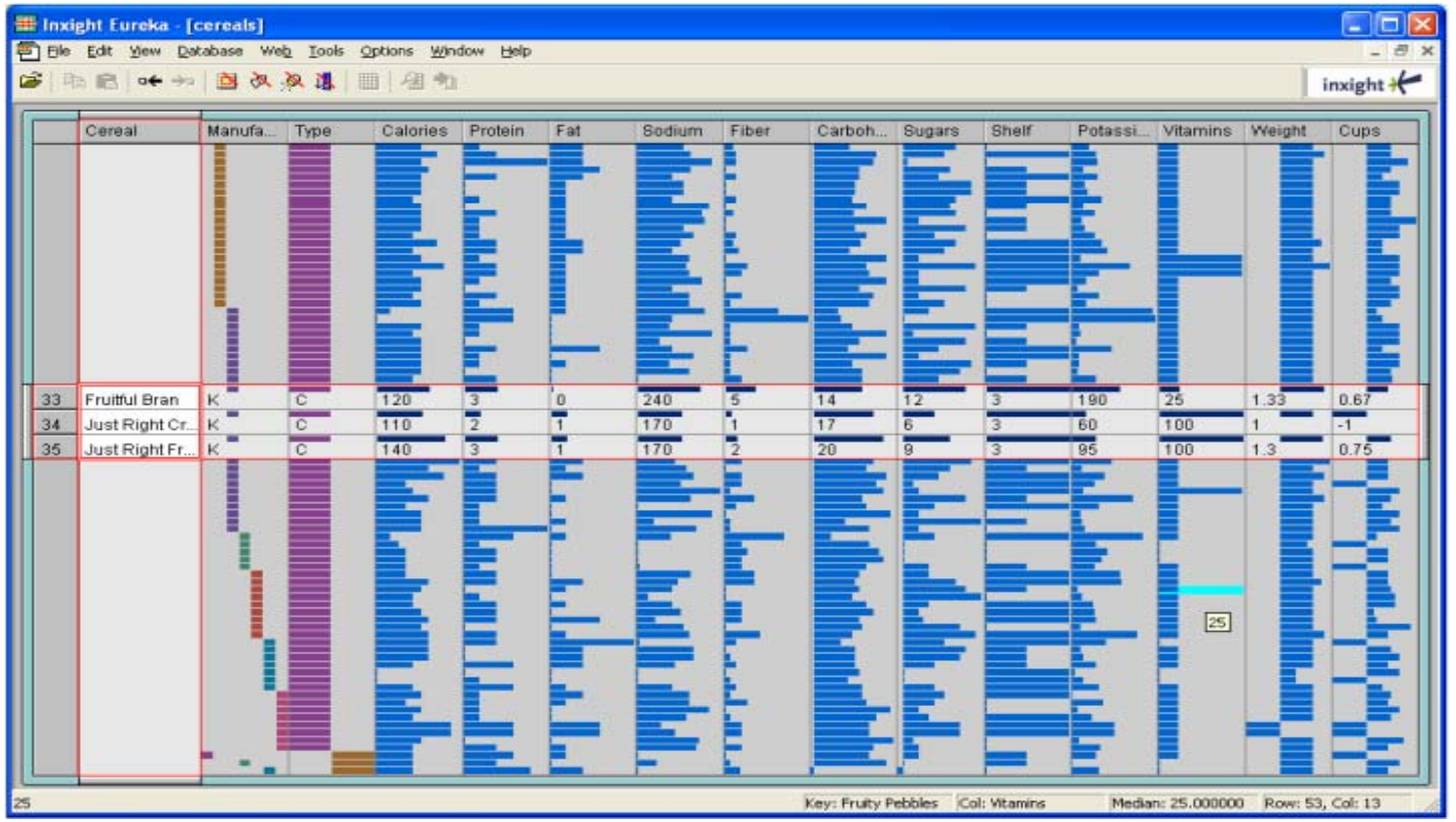
	V_1	V_2	V_3	V_4	V_5	V_6
C_1	4	1	8	3	5	3
C_2	6	3	2	5	1	2
C_3	3	4	5	3	7	5
C_4	6	2	1	4	3	4
C_5	1	2	4	6	3	4



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Visualization approach

- Expanded rows show both the values and the bars
- Nominal values are represented with color and offset



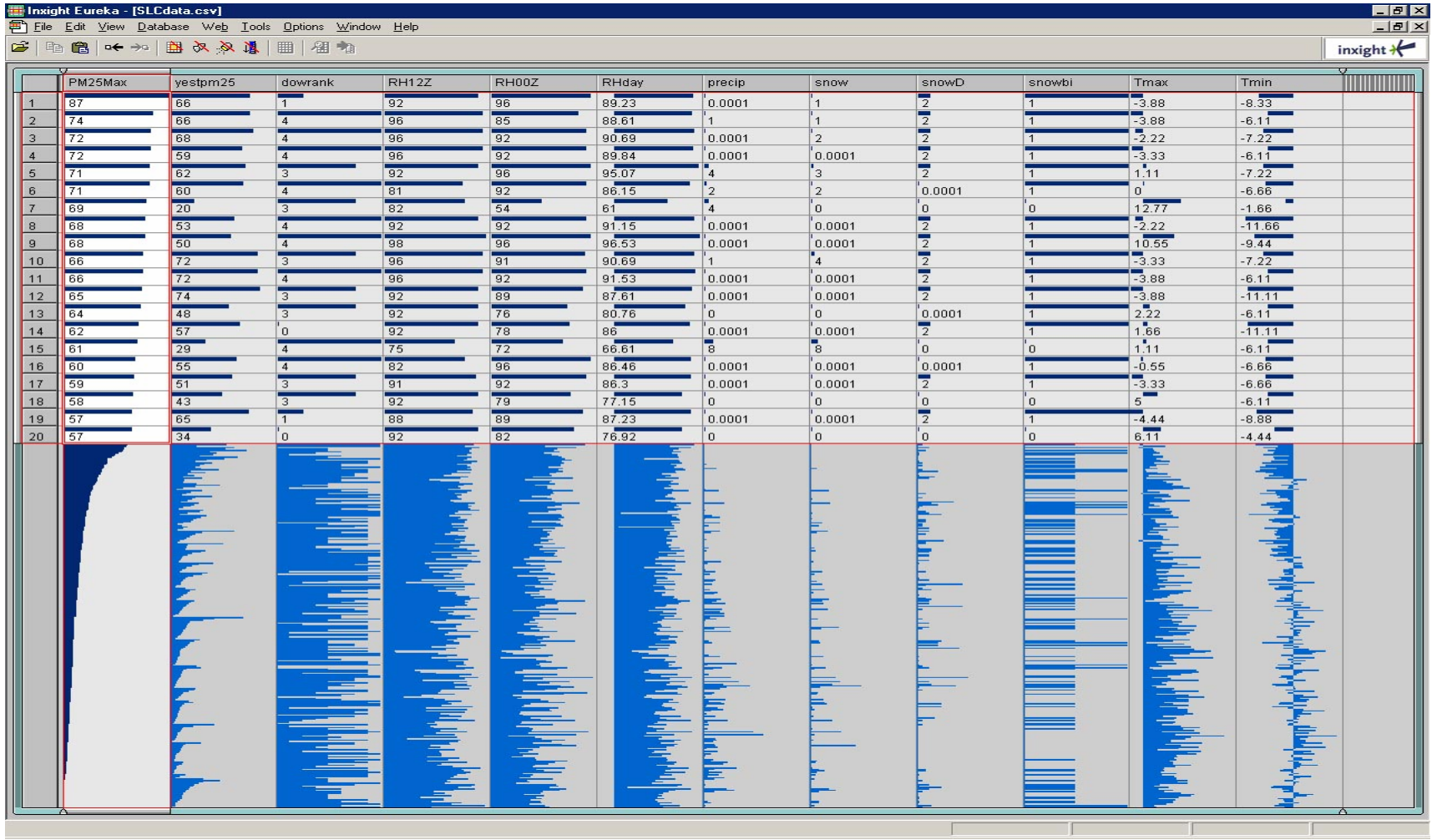
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The lens of table lens

- Table lens boils down to be a powerful focusing tool

Focused rows

Compressed rows

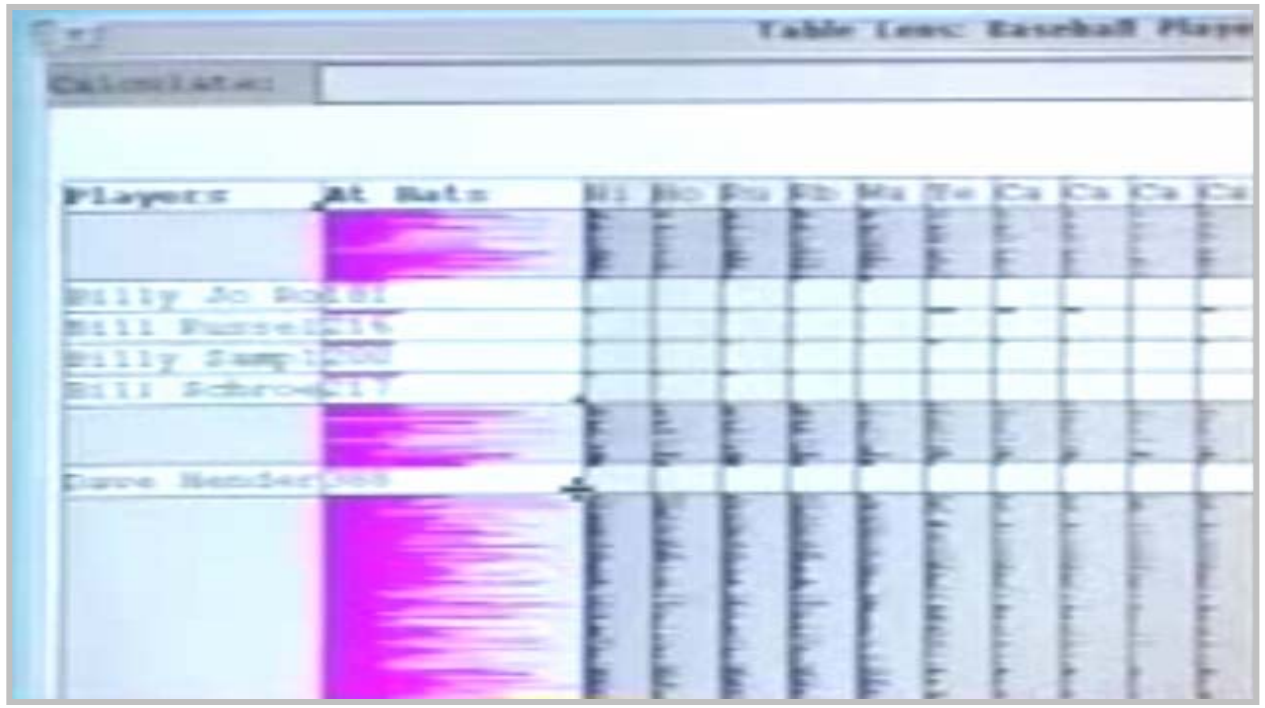


	PM25Max	yestpm25	dowrank	RH12Z	RH00Z	RHday	precip	snow	snowD	snowbi	Tmax	Tmin
1	87	66	1	92	96	89.23	0.0001	1	2	1	-3.88	-8.33
2	74	66	4	96	85	88.61	1	1	2	1	-3.88	-6.11
3	72	68	4	96	92	90.69	0.0001	2	2	1	-2.22	-7.22
4	72	59	4	96	92	89.84	0.0001	0.0001	2	1	-3.33	-6.11
5	71	62	3	92	96	95.07	4	3	2	1	1.11	-7.22
6	71	60	4	81	92	86.15	2	2	0.0001	1	0	-6.66
7	69	20	3	82	54	61	4	0	0	0	12.77	-1.66
8	68	53	4	92	92	91.15	0.0001	0.0001	2	1	-2.22	-11.66
9	68	50	4	98	96	96.53	0.0001	0.0001	2	1	10.55	-9.44
10	66	72	3	96	91	90.69	1	4	2	1	-3.33	-7.22
11	66	72	4	96	92	91.53	0.0001	0.0001	2	1	-3.88	-6.11
12	65	74	3	92	89	87.61	0.0001	0.0001	2	1	-3.88	-11.11
13	64	48	3	92	76	80.76	0	0	0.0001	1	2.22	-6.11
14	62	57	0	92	78	86	0.0001	0.0001	2	1	1.66	-11.11
15	61	29	4	75	72	66.61	8	8	0	0	1.11	-6.11
16	60	55	4	82	96	86.46	0.0001	0.0001	0.0001	1	-0.55	-6.66
17	59	51	3	91	92	86.3	0.0001	0.0001	2	1	-3.33	-6.66
18	58	43	3	92	79	77.15	0	0	0	0	5	-6.11
19	57	65	1	88	89	87.23	0.0001	0.0001	2	1	-4.44	-8.88
20	57	34	0	92	82	76.92	0	0	0	0	6.11	-4.44

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Table lens demo

- Table lenses were commercialized by Inxight Software, a Xerox PARC spin off
- Watch Rao and Card video



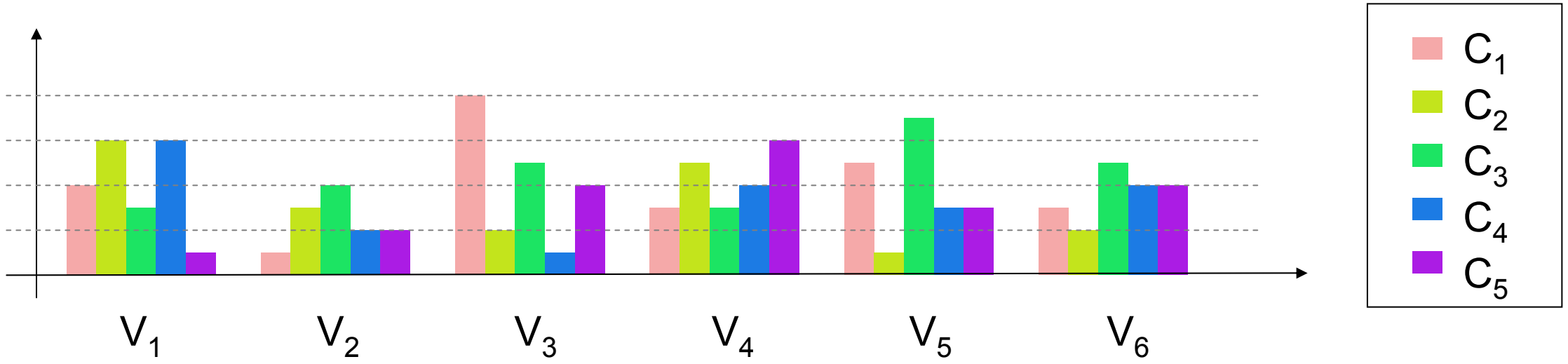
Video at http://www.youtube.com/watch?v=qWqTrRAC52U

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Multiple bars

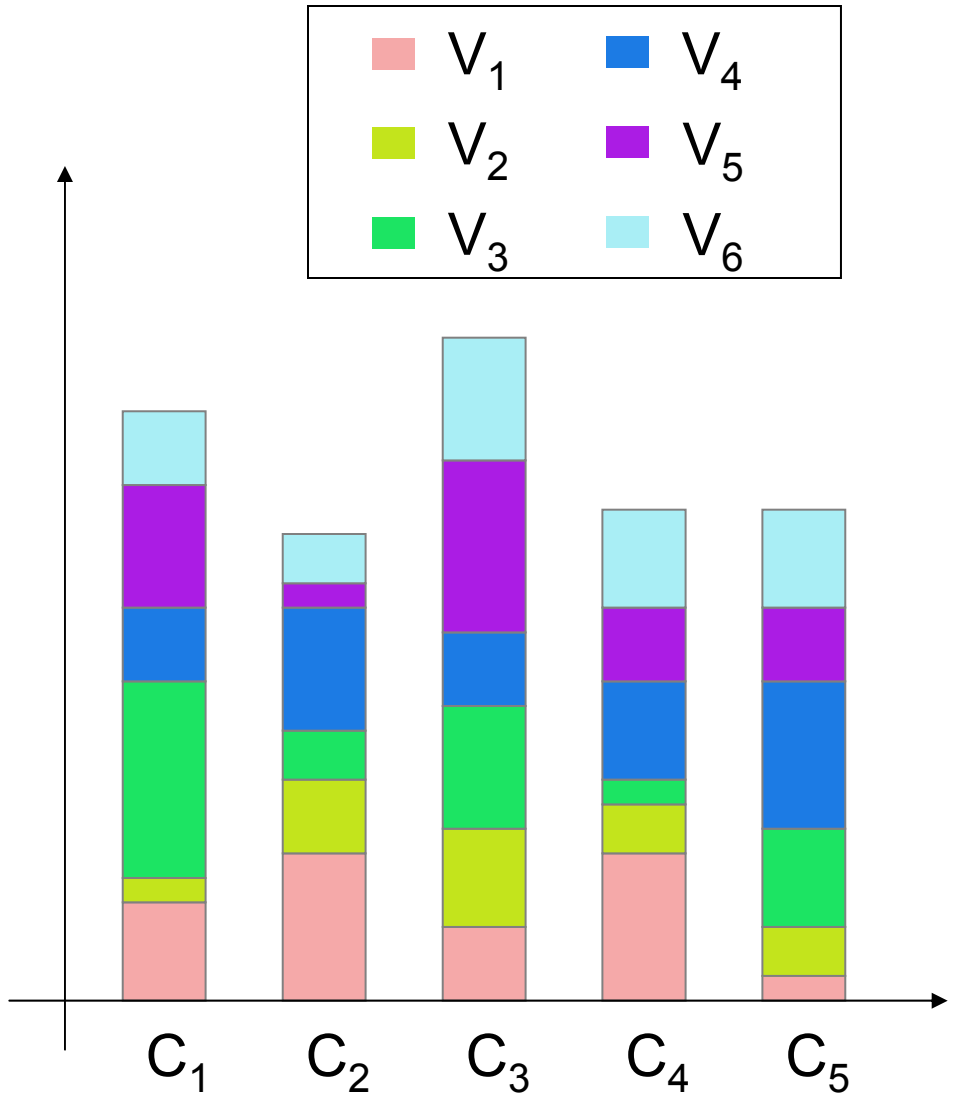
- Bar charts where the bars are grouped by variables and colored by data case
  - also called “grouped bars”

	V_1	V_2	V_3	V_4	V_5	V_6
C_1	4	1	8	3	5	3
C_2	6	3	2	5	1	2
C_3	3	4	5	3	7	5
C_4	6	2	1	4	3	4
C_5	1	2	4	6	3	4



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Stacked bars

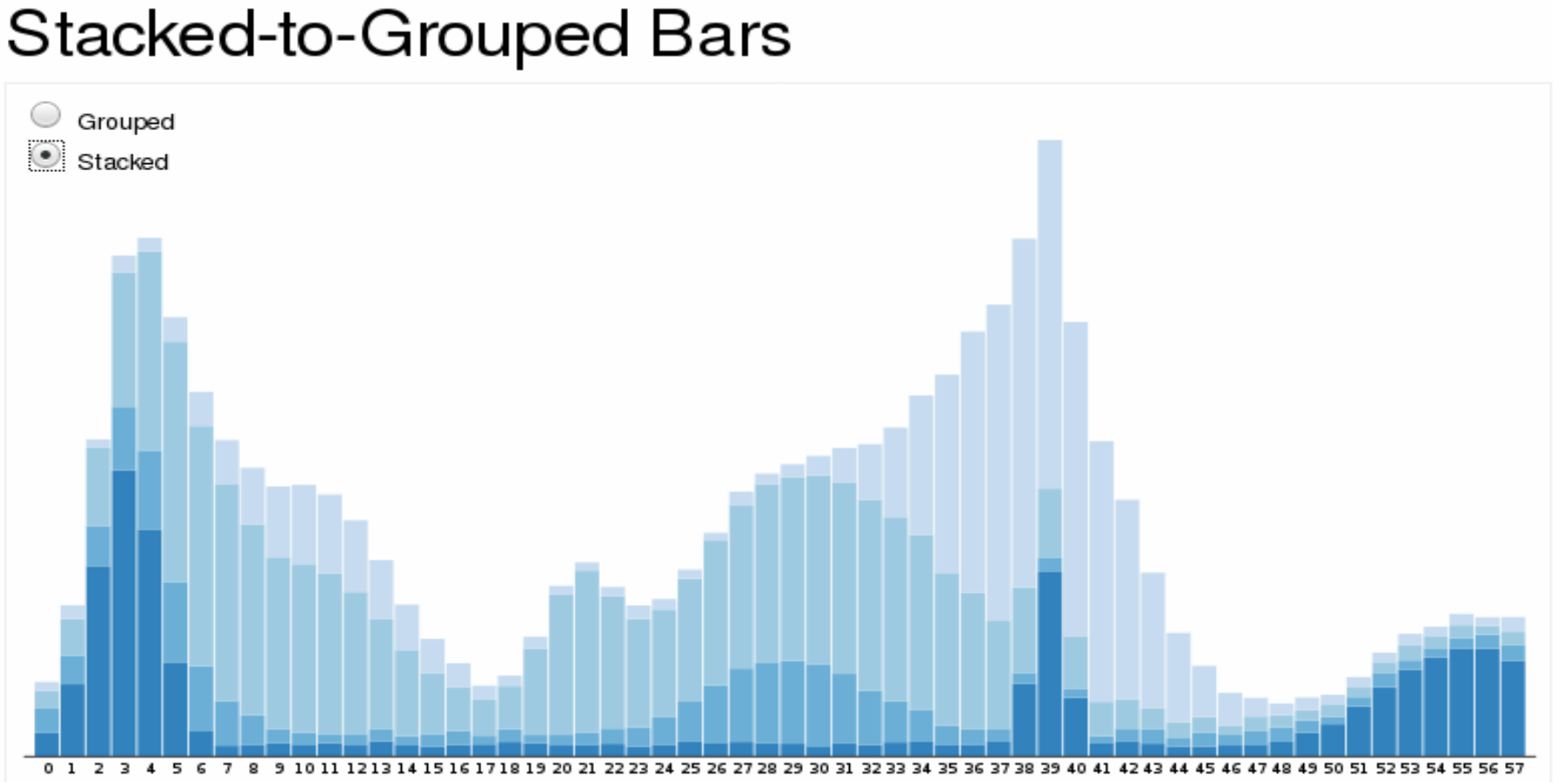


	V_1	V_2	V_3	V_4	V_5	V_6
C_1	4	1	8	3	5	3
C_2	6	3	2	5	1	2
C_3	3	4	5	3	7	5
C_4	6	2	1	4	3	4
C_5	1	2	4	6	3	4

- Not a clear representation of the data
  - unless the sum of the variables is significant

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Stacked bars example



http://bl.ocks.org/mbostock/3943967

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Motivation for using spineplots

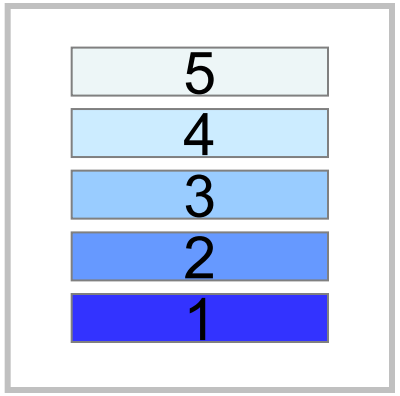
- Example dataset: 1978 car data
  - data and figures from [Cox 08]
- Variables
  - nationality
    - domestic or foreign cars
- Repair records
  - 1 = best
  - 5 = worst
- Model

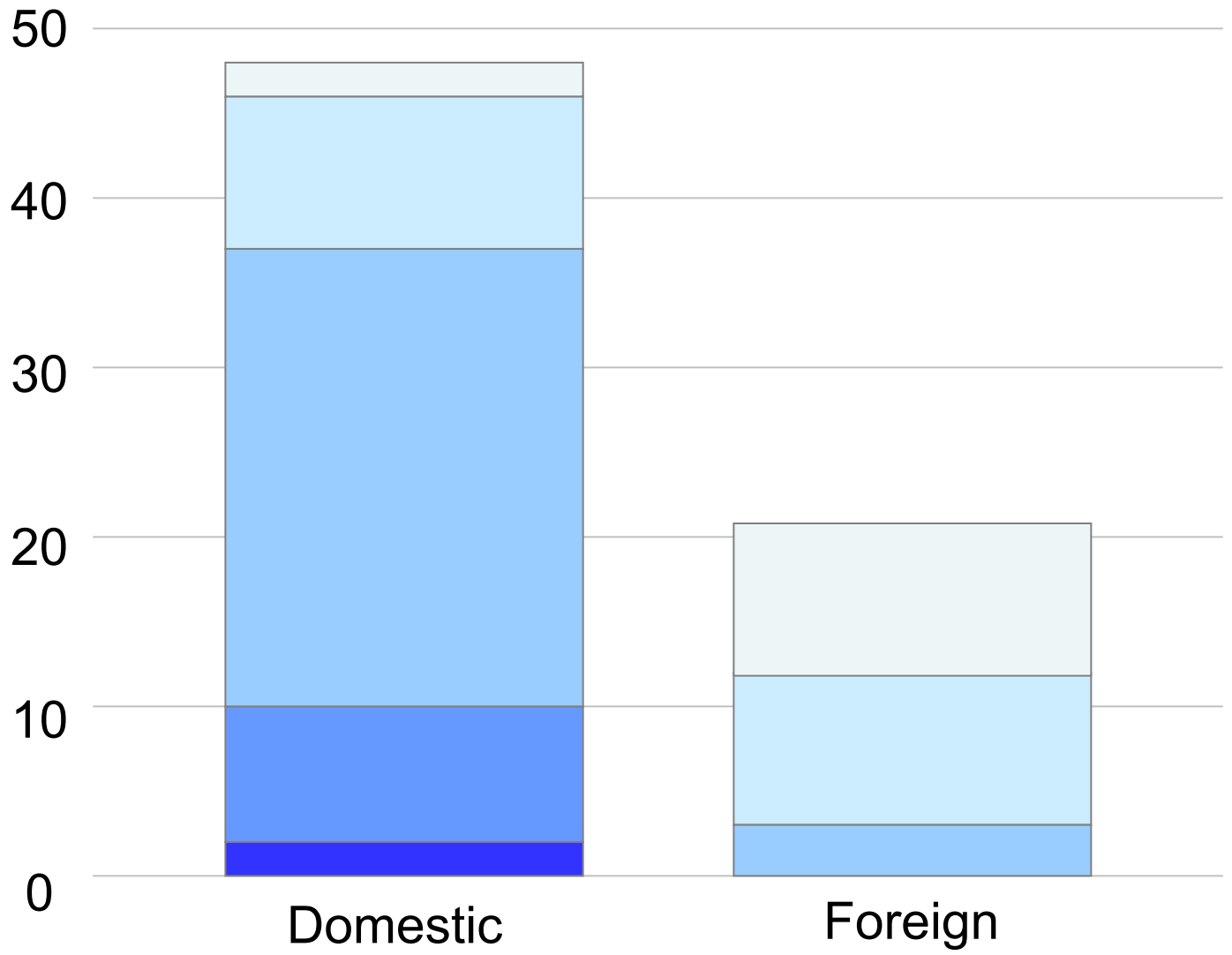
Nationality	Repair records	# models
domestic	1	2
domestic	2	8
domestic	3	27
domestic	4	9
domestic	5	2
foreign	1	0
foreign	2	0
foreign	3	3
foreign	4	9
foreign	5	9

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Representing data with stacked bars

- Conveys absolute values but not percentages

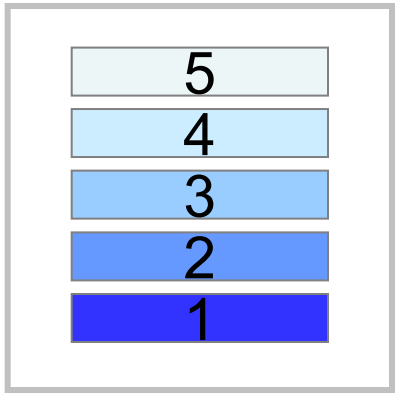


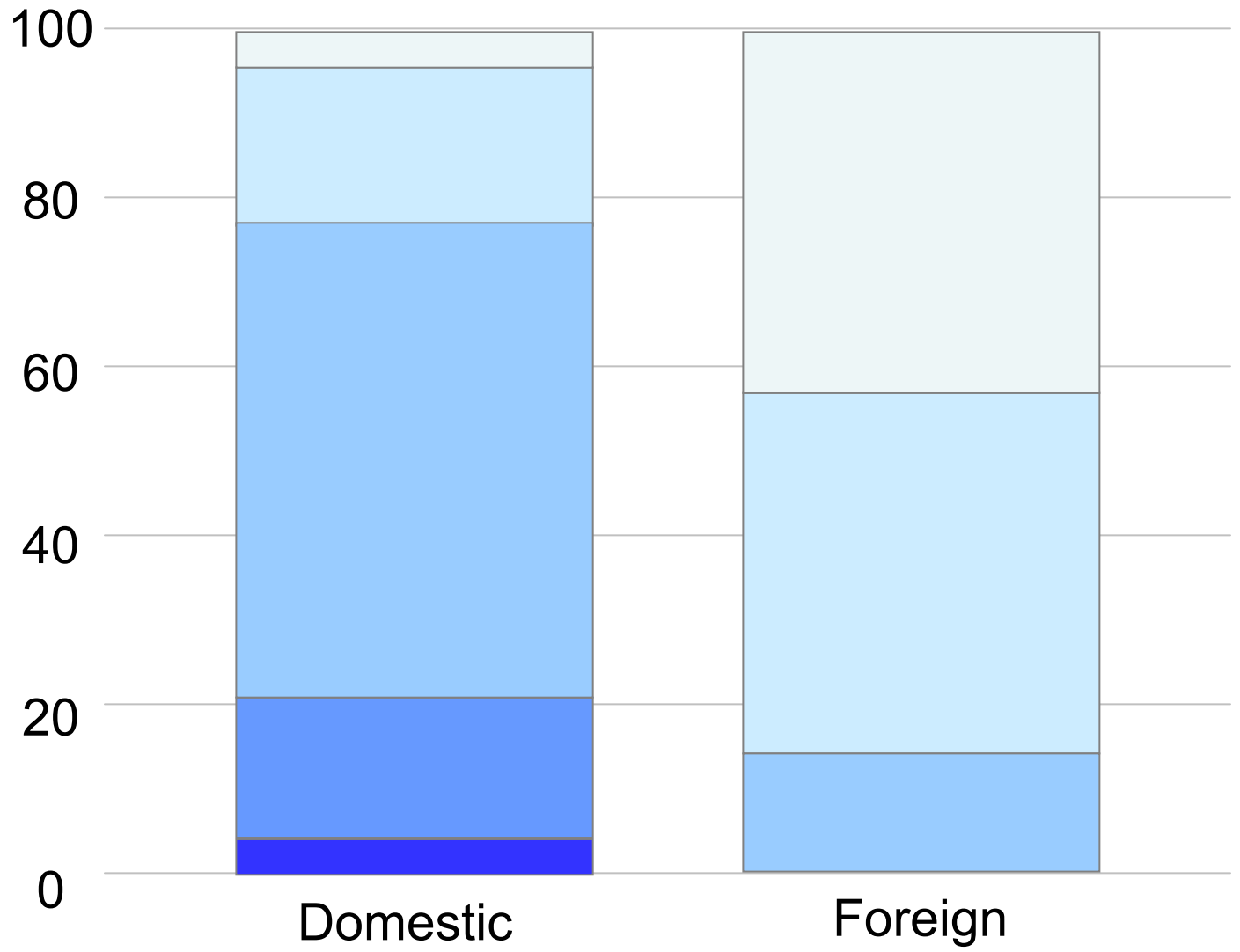


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Stacked bars with percentages

- Conveys percentages but not domestic vs foreign proportion

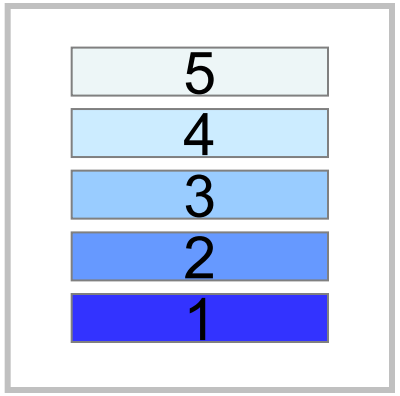


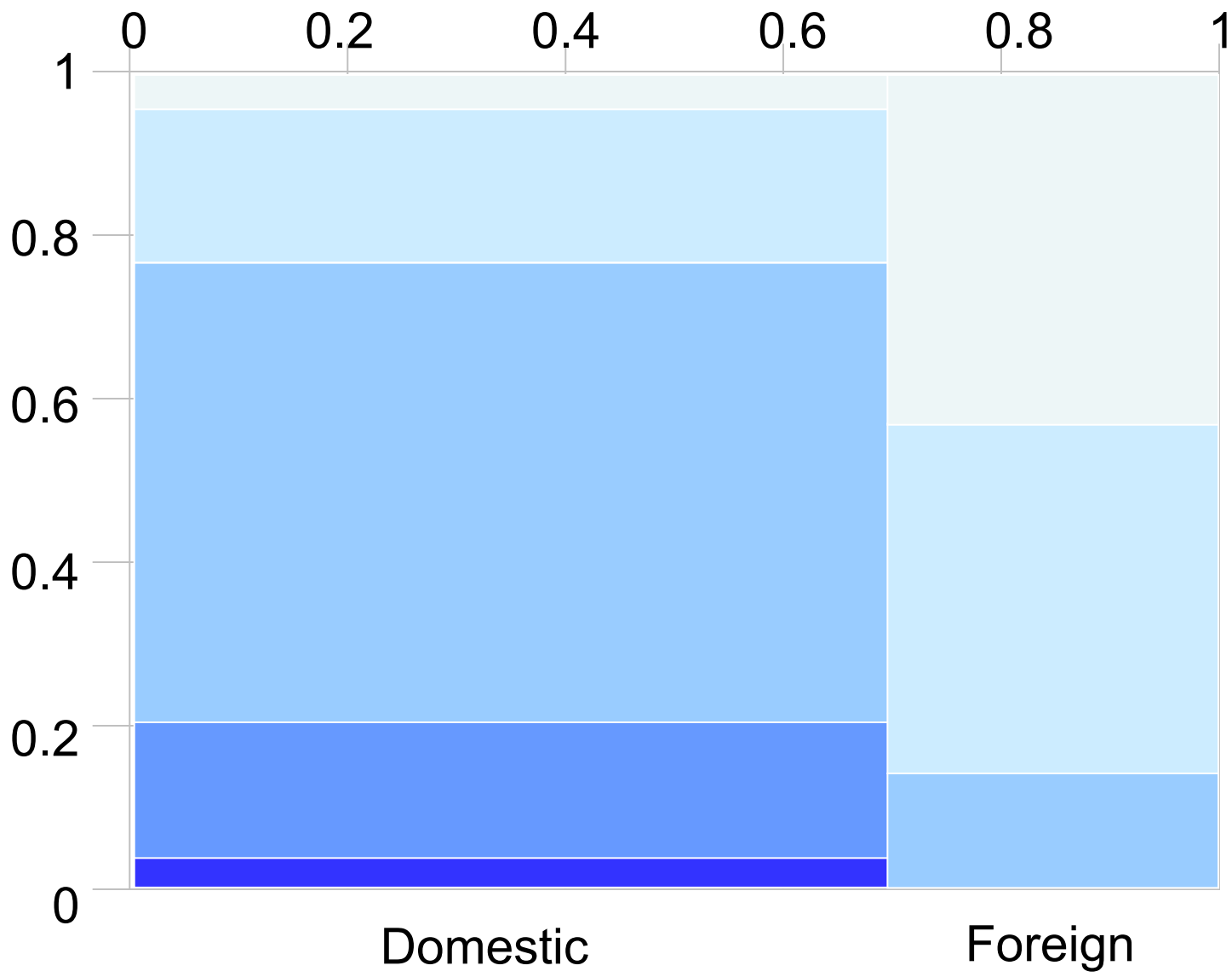


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Spineplot

- Conveys both percentages and domestic vs foreign proportion





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Generalization: mosaic plots

- Example dataset: Titanic survivors
  - data from Wikipedia “Mosaic plots”

Gender	Survived	1° class	2° class	3° class	Crew
Male	No	118	154	422	670
Male	Yes	62	25	88	192
Female	No	4	13	106	3
Female	Yes	141	93	90	20

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Generalization: mosaic plots

- Data transformation

Gender	Class	Survived	Number
Male	1°	yes	62
Male	1°	no	118
Male	2°	yes	25
Male	2°	no	154
Male	3°	yes	88
Male	3°	no	422
Male	Crew	yes	192
Male	Crew	no	670

Gender	Class	Survived	Number
Female	1°	yes	141
Female	1°	no	4
Female	2°	yes	93
Female	2°	no	13
Female	3°	yes	90
Female	3°	no	106
Female	Crew	yes	20
Female	Crew	no	3

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Generalization: mosaic plots

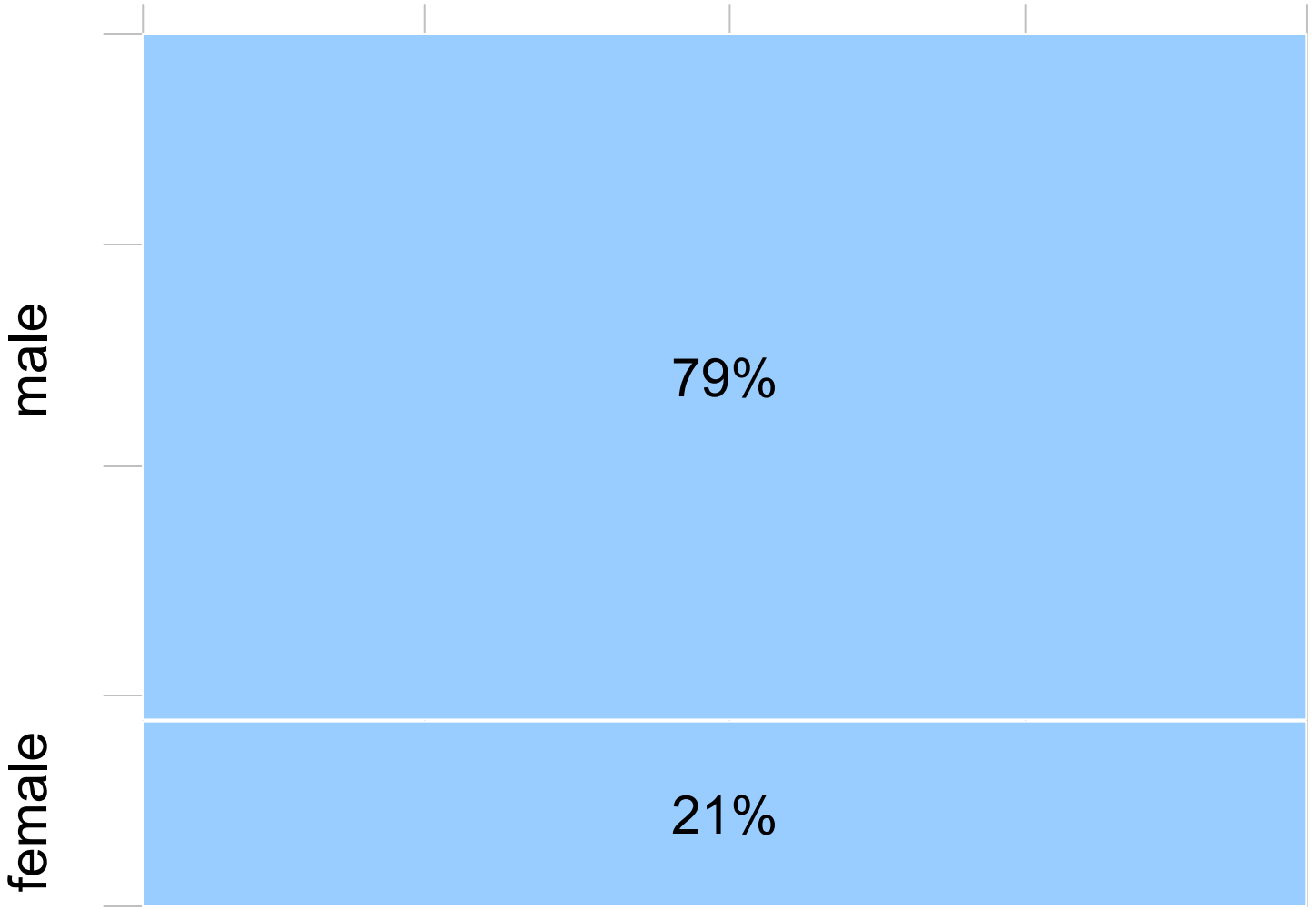
- Order the independent variables
- Alternatively assign vertical/horizontal axes

Order	Variable	Axis
1	Gender	Vertical
2	Class	Horizontal
3	Survived	Vertical

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Creating a mosaic plot: 1st variable

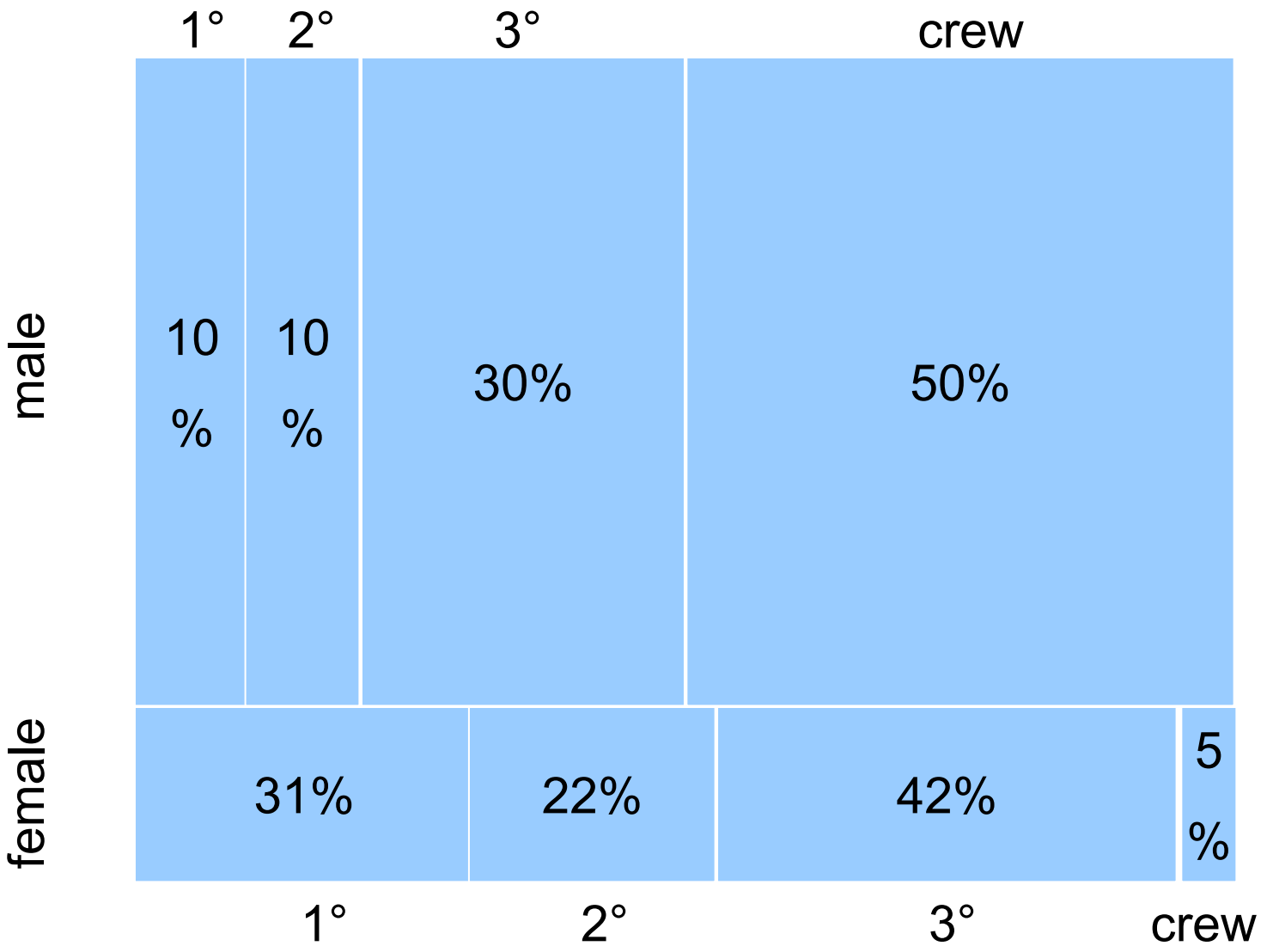
- 1st variable
  - gender
  - vertical axis



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Creating a mosaic plot: 2nd variable

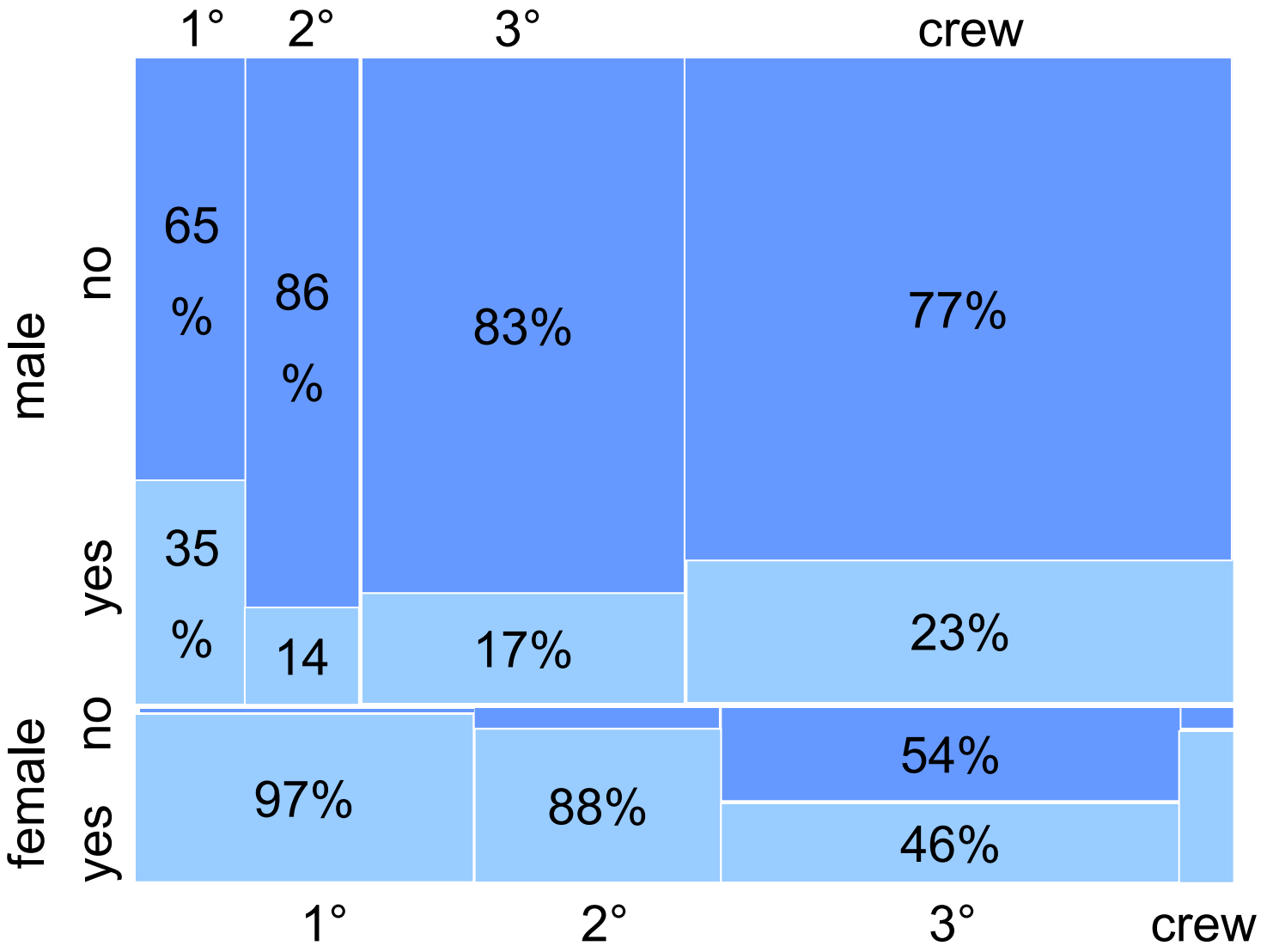
- 1st variable
  - gender
  - vertical axis
- 2nd variable
  - class
  - horizontal axis



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Creating a mosaic plot: 3rd variable

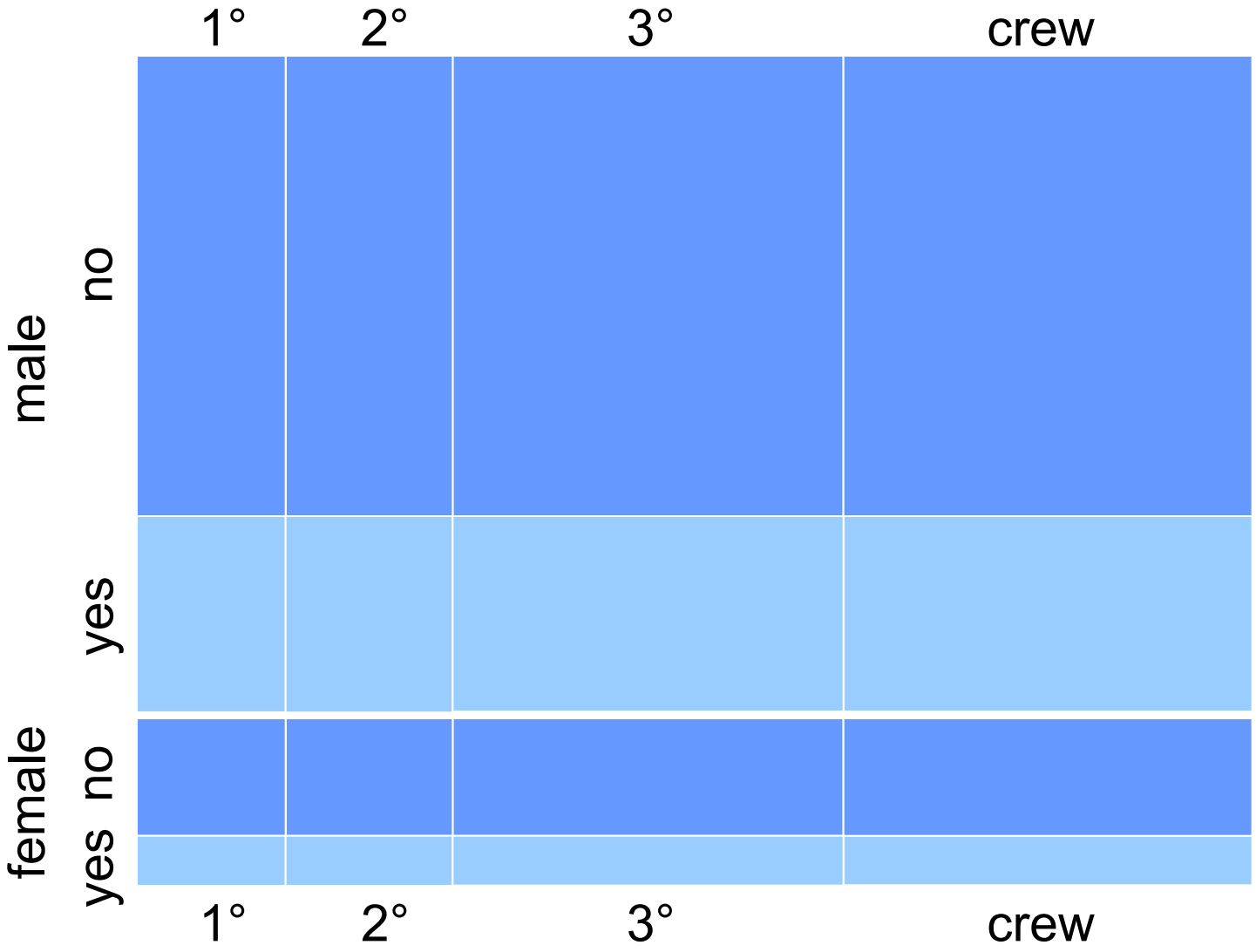
- 1st variable
  - gender
  - vertical axis
- 2nd variable
  - class
  - horizontal axis
- 3rd variable
  - survived
  - vertical axis



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Mosaic plots: independent variables

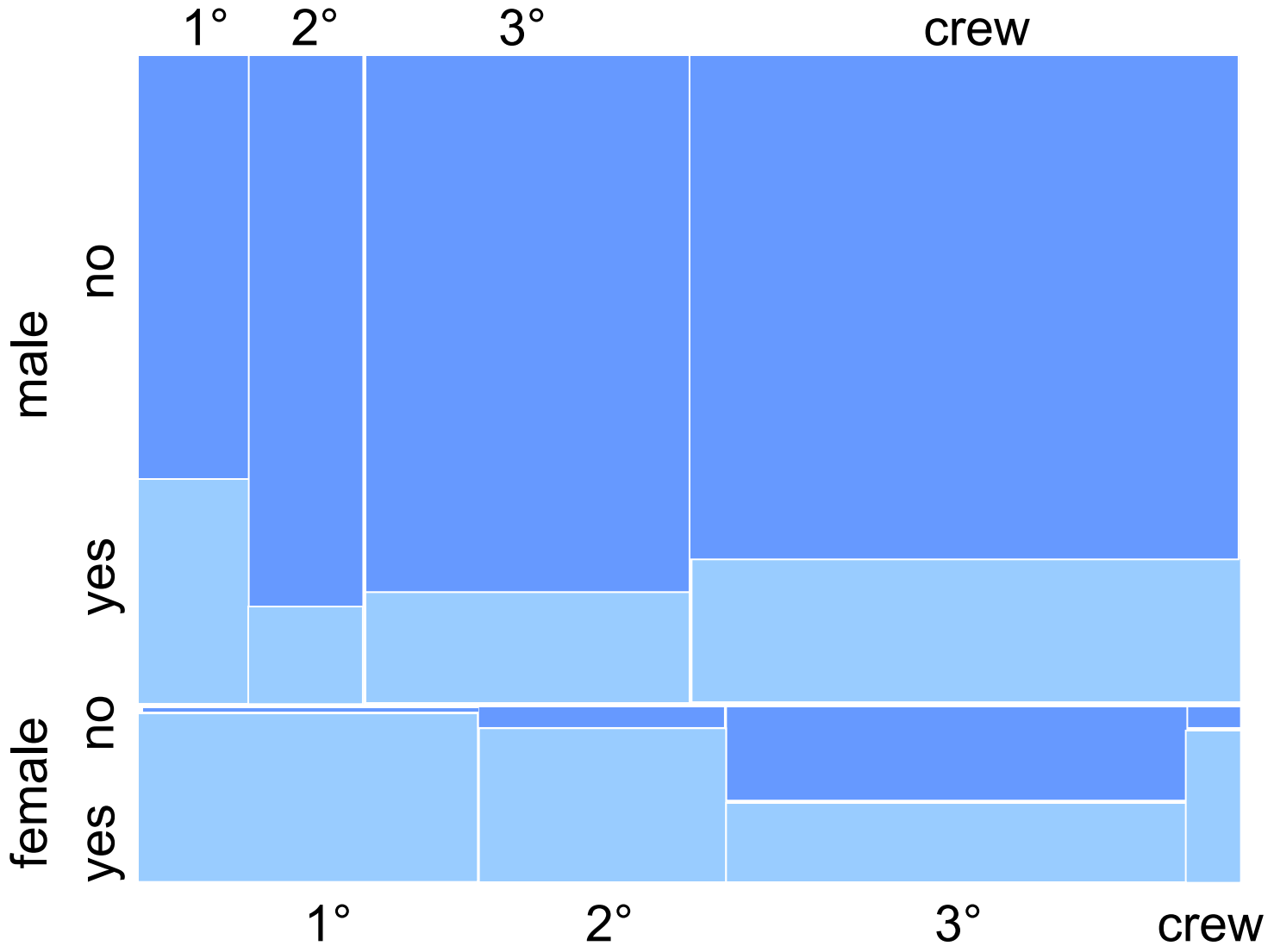
- Example of a dataset where variables have no relationship
  - gender does not imply class
  - gender and class do not impact survival



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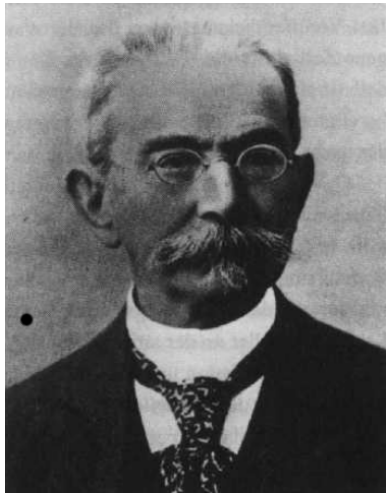
Final mosaic plot

- Pros
  - spacefilling
  - shows variable dependency
- Cons
  - difficult to extend to more than a few variables

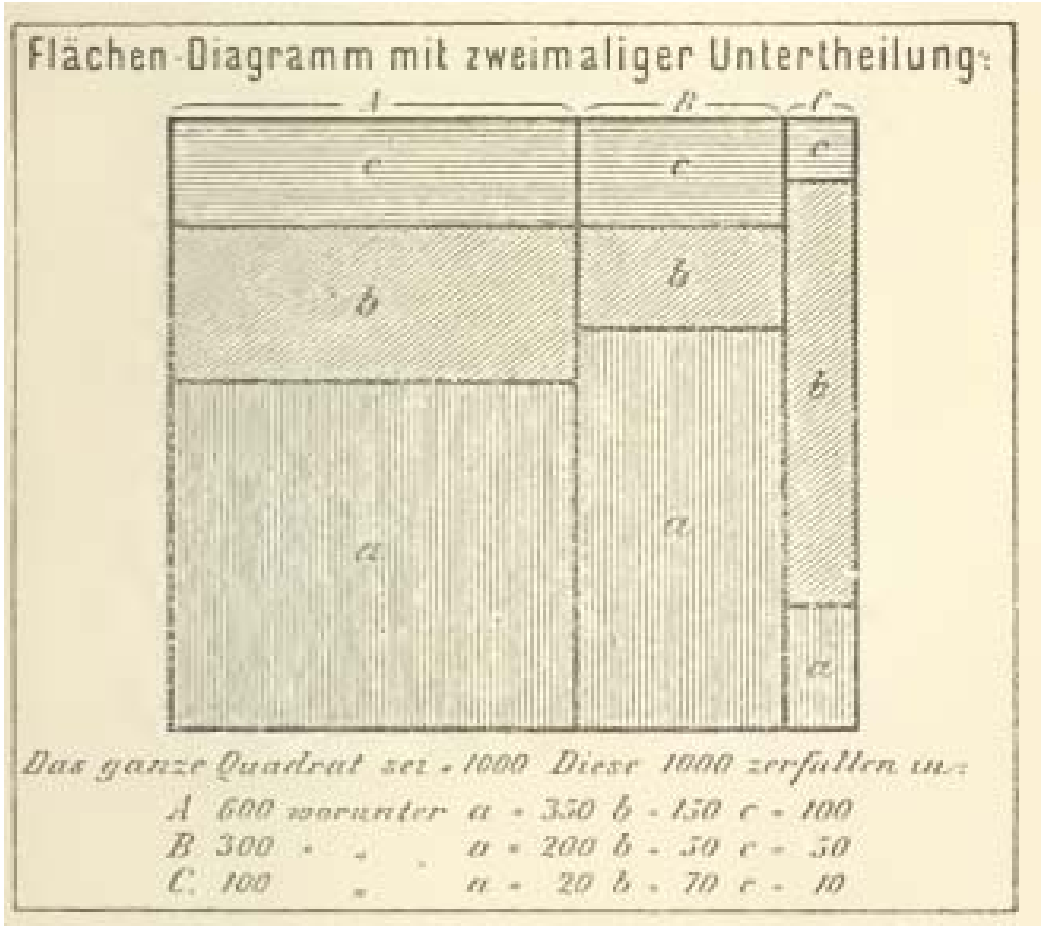


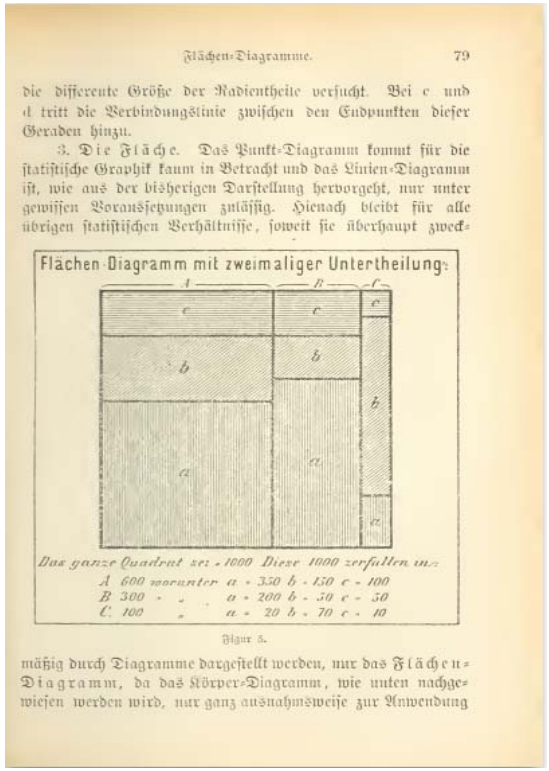
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Mosaic plots history



- Introduced by Georg von Mayr in 1877




Flächen-Diagramme. 79

die differente Größe der Radientheile versucht. Bei c und d tritt die Verbindungslinie zwischen den Endpunkten dieser Geraden hinzu.

3. Die Fläche. Das Punkt-Diagramm kommt für die statistische Graphik kaum in Betracht und das Linien-Diagramm ist, wie aus der bisherigen Darstellung hervorgeht, nur unter gewissen Voraussetzungen zulässig. Hienach bleibt für alle übrigen statistischen Verhältnisse, soweit sie überhaupt zweck-

Flächen-Diagramm mit zweimaliger Untertheilung.

Das ganze Quadrat sei = 1000. Diese 1000 zerfallen in:
A 600 worunter a = 350 b = 150 c = 100
B 300 " " a = 200 b = 50 c = 50
C. 100 " a = 20 b = 70 c = 10

Figur 5.

mäßig durch Diagramme dargestellt werden, nur das Flächen-Diagramm, da das Körper-Diagramm, wie unten nachgewiesen werden wird, nur ganz ausnahmsweise zur Anwendung

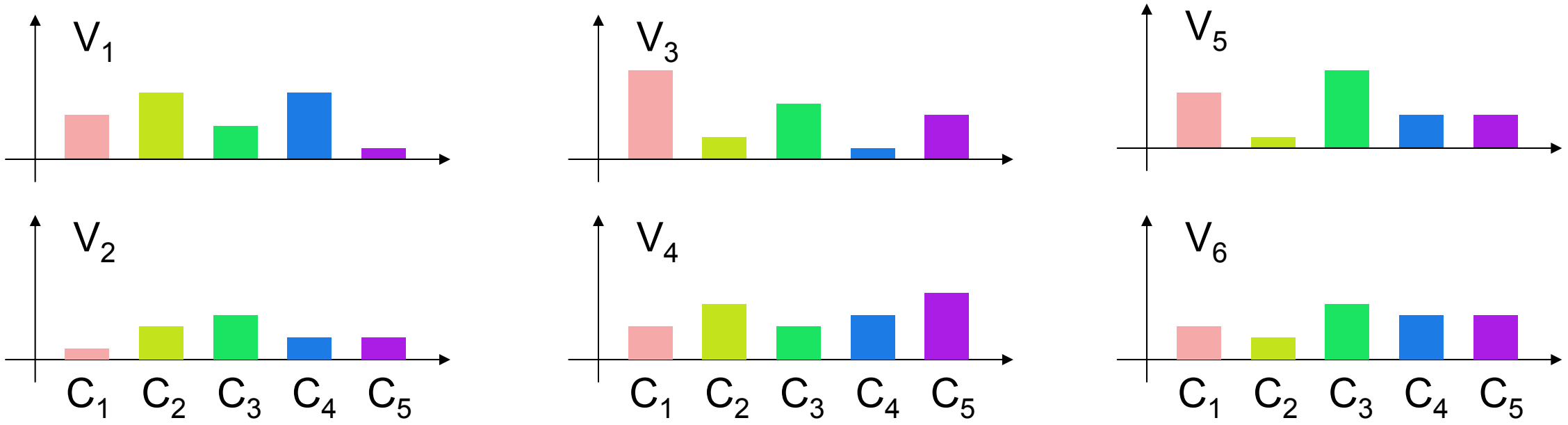
page 78 of “Die Gesetzmässigkeit im Gesellschaftsleben statistische Studien“

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Small multiples

- Strategy
  - each variable has its own display
- Presenting a matrix of smaller charts is also called “multiple views”

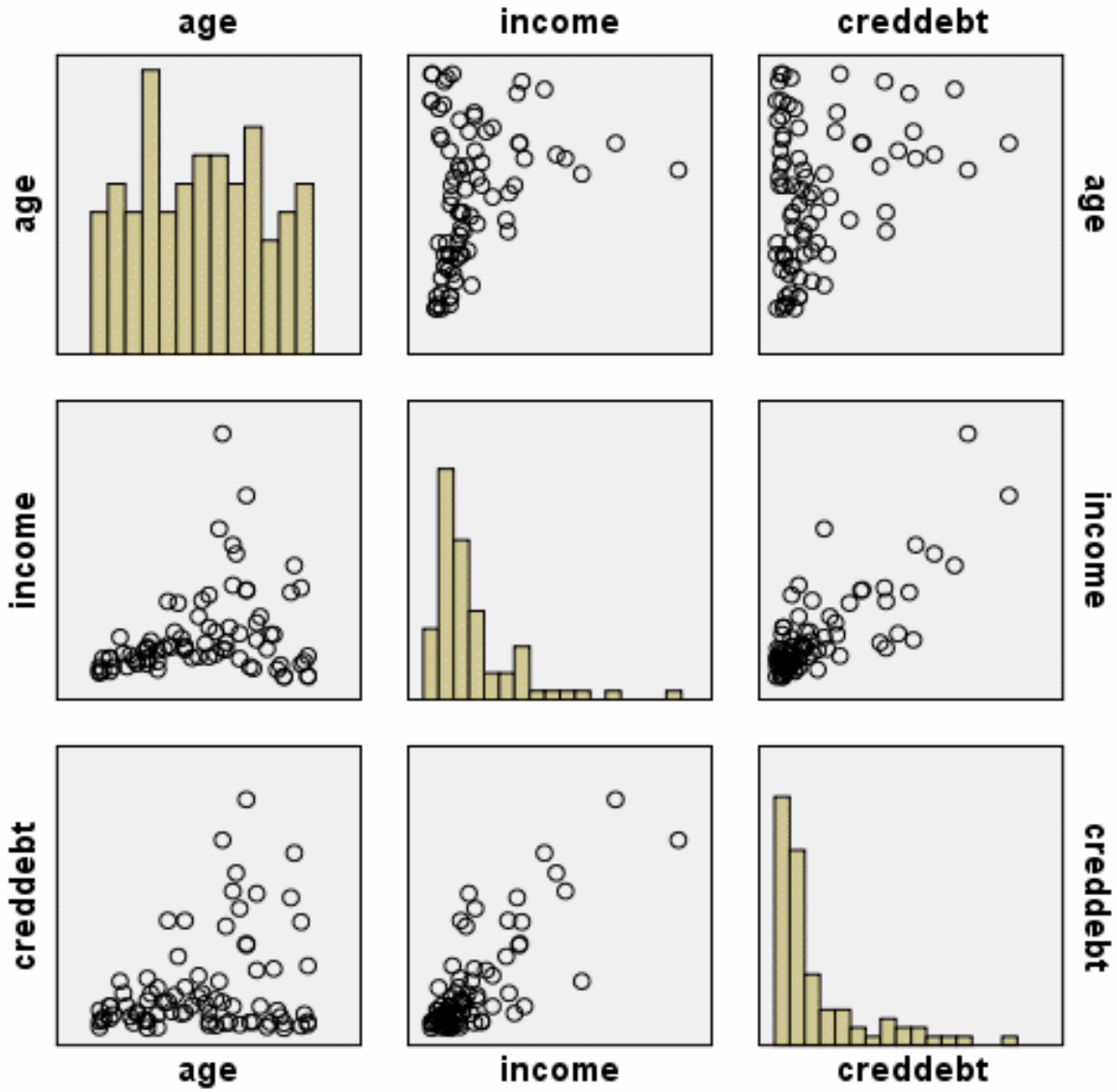
	V_1	V_2	V_3	V_4	V_5	V_6
C_1	4	1	8	3	5	3
C_2	6	3	2	5	1	2
C_3	3	4	5	3	7	5
C_4	6	2	1	4	3	4
C_5	1	2	4	6	3	4



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Scatterplot matrix (SPLOM)

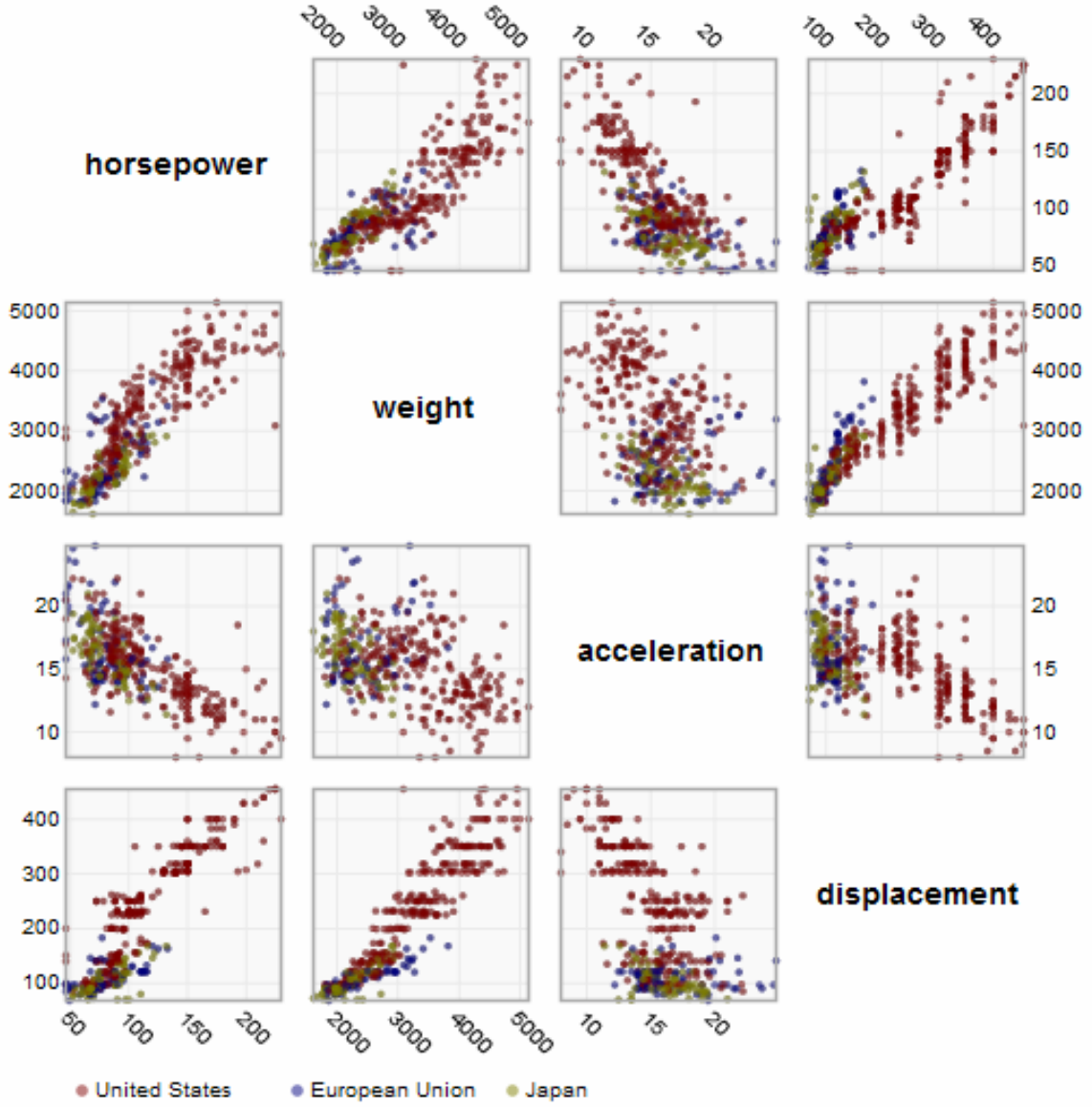
- Also called SPLOM
  - stands for *Scatter PLOt Matrix*
- Given n variables represent each possible pair of them in their own 2D scatterplot
  - the diagonal is usually devoted to the variable distribution



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A bigger example

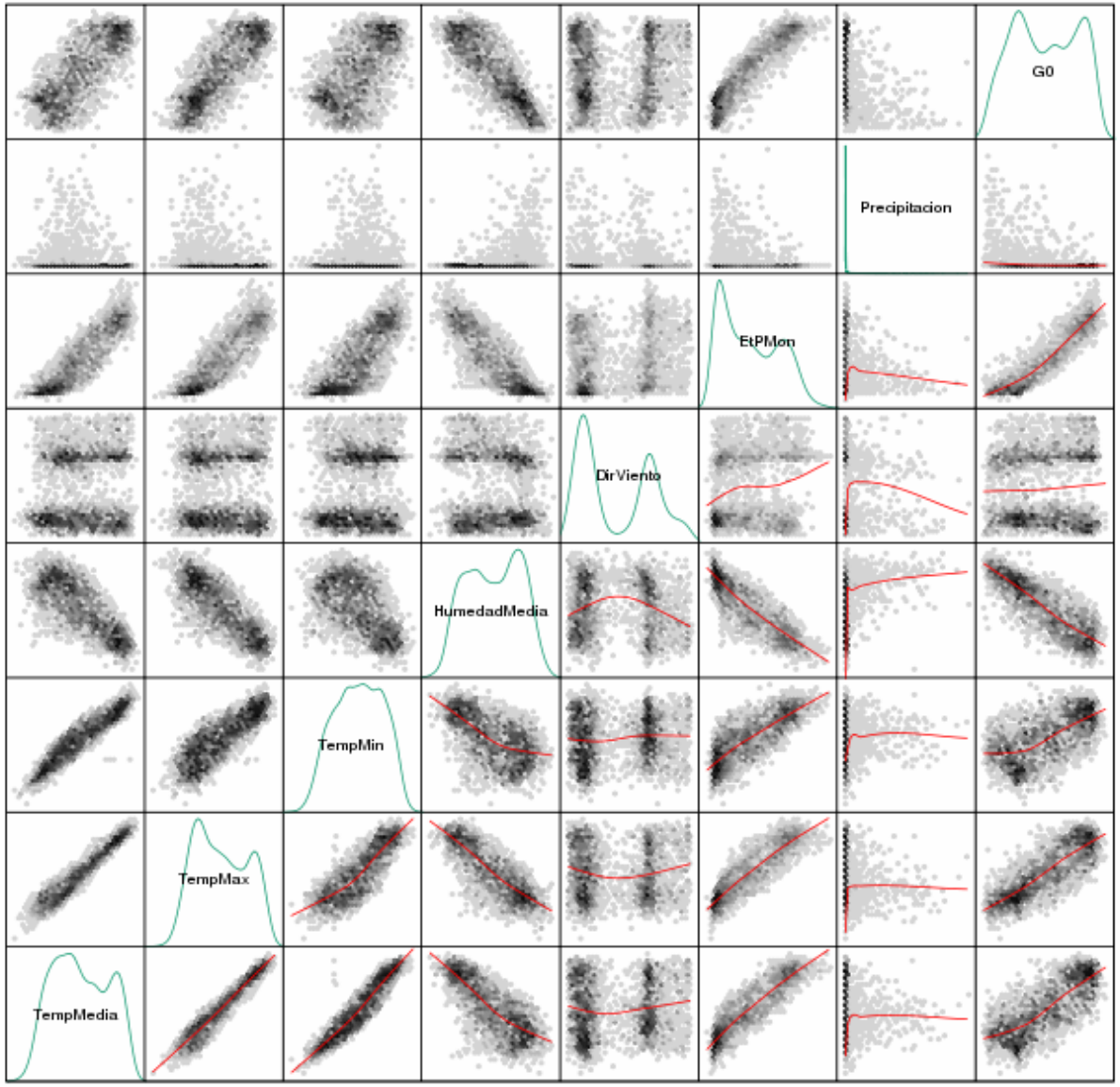
- A bigger example
  - from [Heer *et al.* 14]



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Huge example

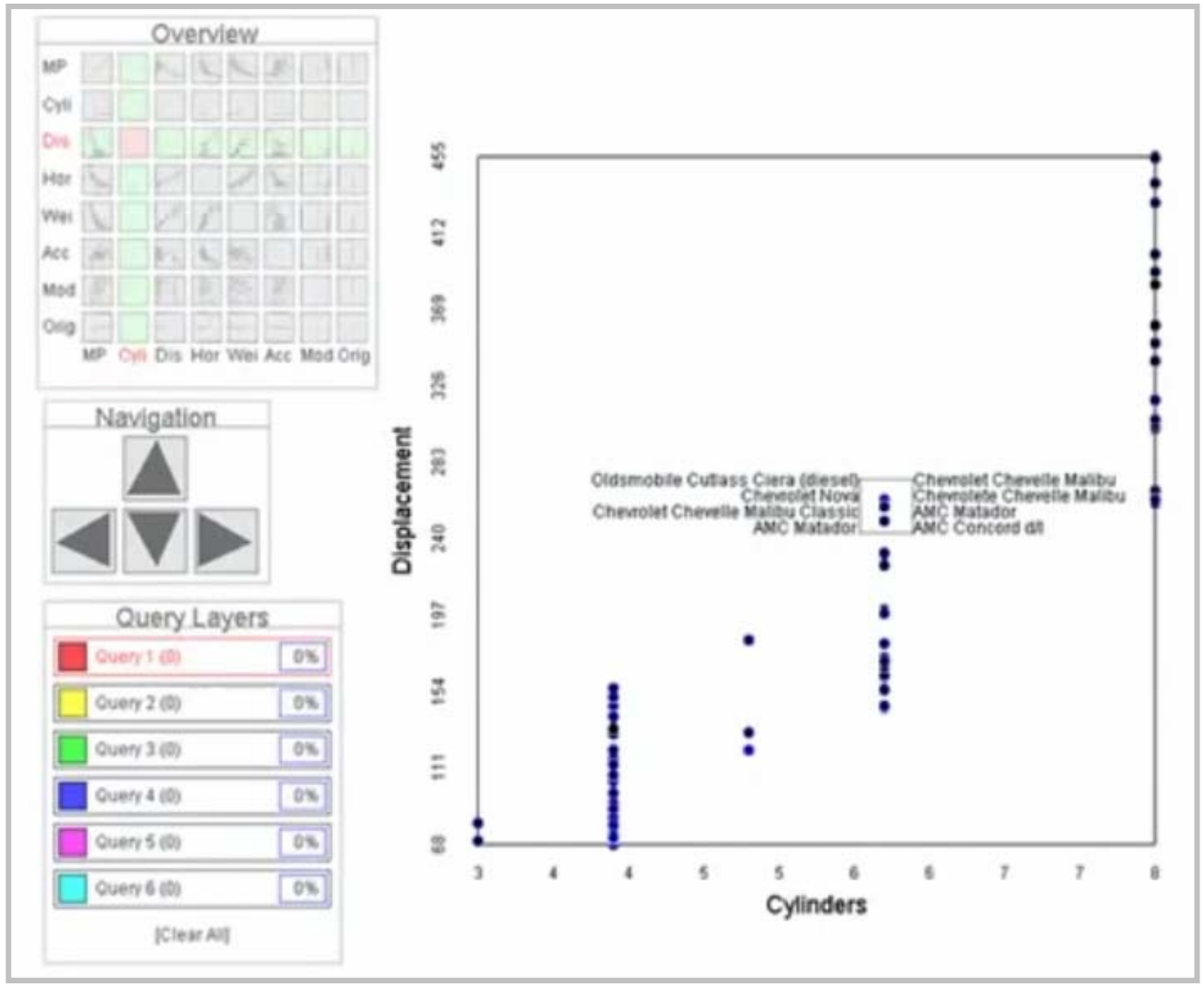
- An 8×8 scatterplot matrix
  - from [Perpiñán 11]



Scatter Plot Matrix

Navigating a scatterplot matrix

- ScatterDice
  - video from [Elmqvist *et al.* 08]

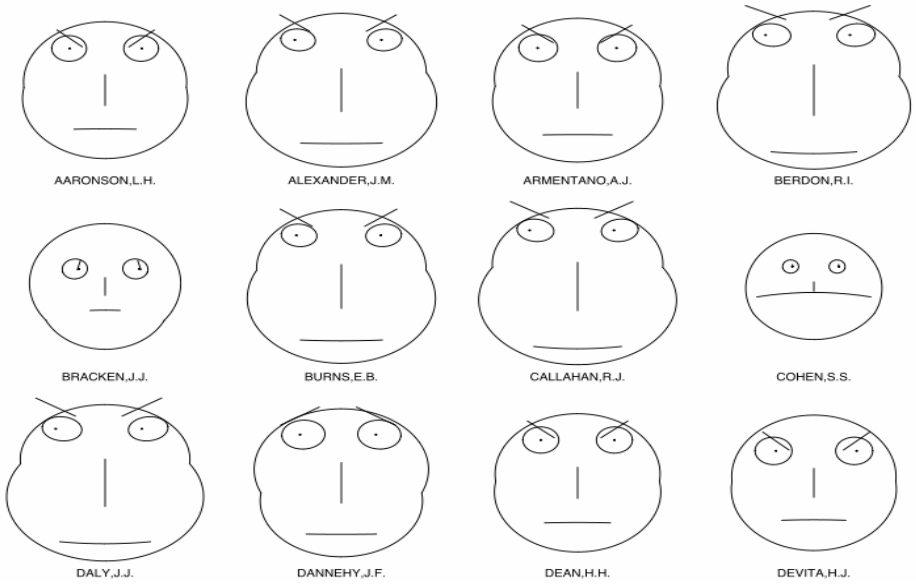


Video at http://www.youtube.com/watch?v=2bYIRcO-gwg

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Chernoff faces

- Visualization technique introduced by Herman Chernoff in 1973
- Humans easily recognize faces and notice small changes without difficulty
- Use facial features to encode several variable values
  - the way in which variables are mapped to the features should be carefully chosen



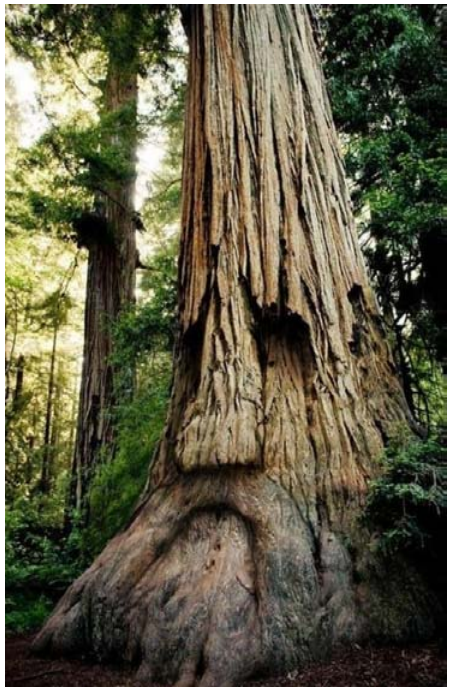
Evaluation of US judges (Wikipedia)

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A motivation for Chernoff faces

- *Pareidolia* is the incorrect perception of a stimulus as an object, pattern or meaning known to the observer
  - examples include seeing shapes in clouds, seeing faces in inanimate objects or abstract patterns, or hearing hidden messages in music



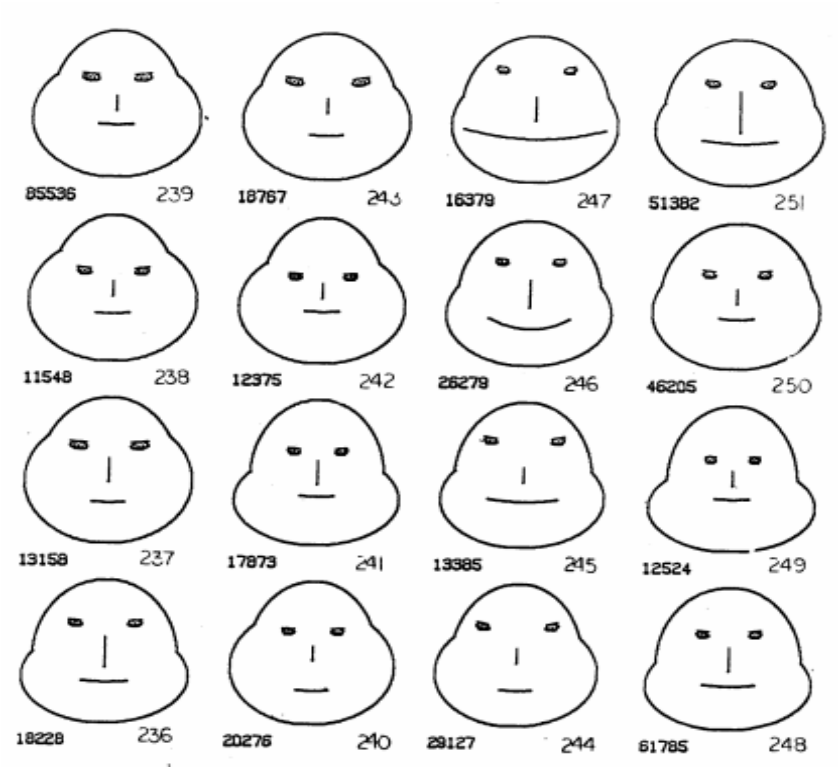






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Chernoff faces

- Chernoff used eighteen facial features

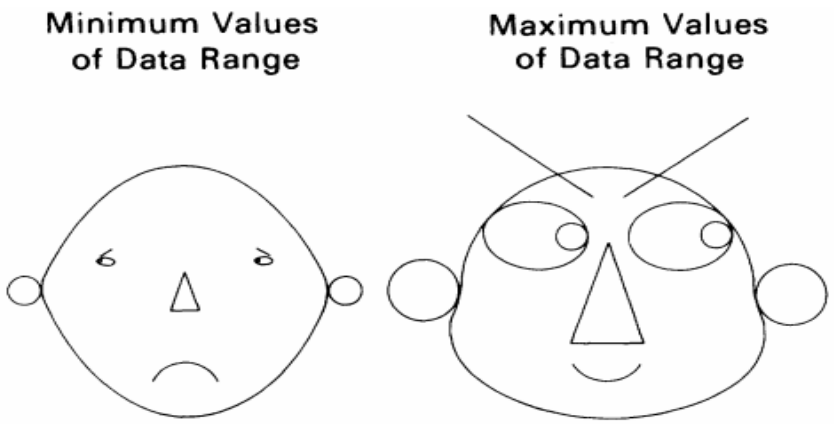


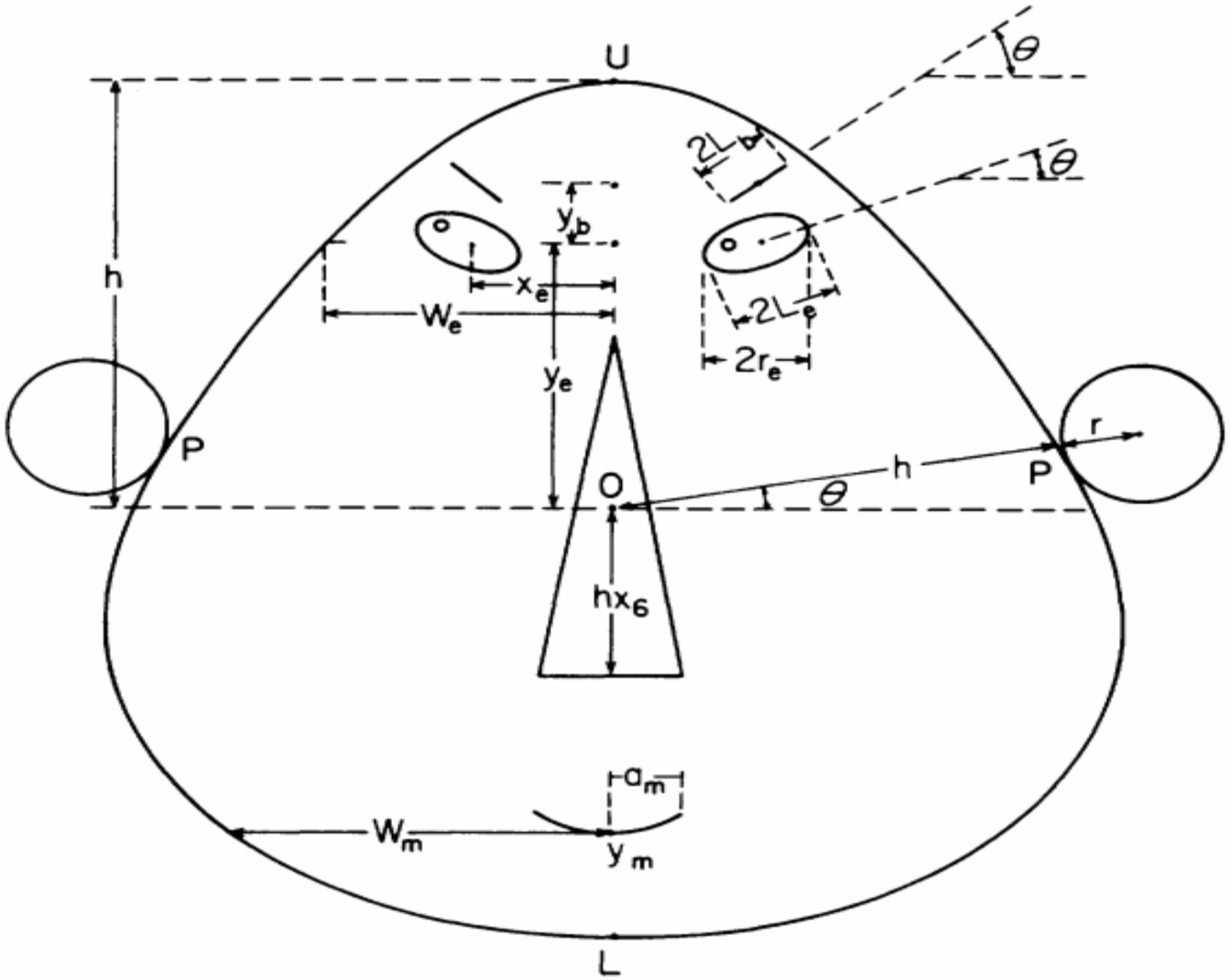
x_1 radius to corner of face, |OP|
x_2 angle of OP to horizontal
x_3 vertical size of face, |OU|
x_4 eccentricity of upper face
x_5 eccentricity of lower face
x_6 length of nose
x_7 vertical position of mouth
x_8 curvature of mouth
x_9 width of mouth
x_{10} vertical position of eyes
x_{11} separation of eyes
x_{12} slant of eyes
x_{13} eccentricity of eyes
x_{14} size of eyes
x_{15} position of pupils
x_{16} vertical position of eyebrows
x_{17} slant of eyebrows
x_{18} size of eyebrows

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Chernoff faces

- In [Huff *et al.* 81] nineteen facial features are controlled





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A financial example

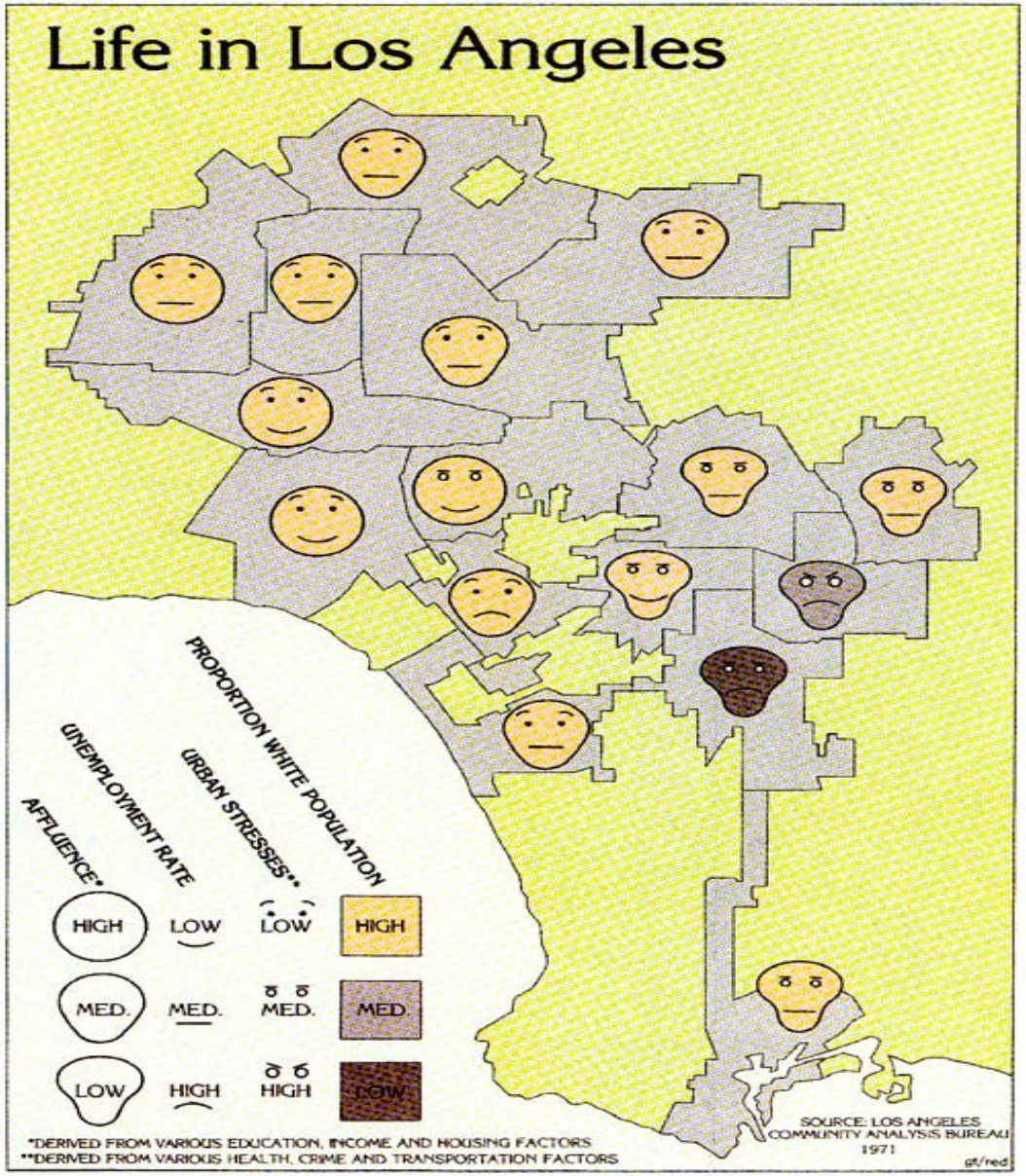
FEDERAL					
Date / Dimensions	Year to Failure 5	4	3	2	1
1. Return on Assets	0.10	0.11	0.06	0.03	−0.16
2. Debt Service	3.66	3.79	1.55	0.78	−14.11
3. Cash Flows	1.53	1.48	1.39	1.35	0.94
4. Capitalization	0.22	0.20	0.18	0.16	−0.02
5. Current Ratio	71.40	89.10	97.85	96.80	58.21
6. Cash Turnover	24.03	25.92	25.62	27.40	71.26
7. Receivables Turnover	5.25	4.46	4.26	4.36	9.56
8. Inventory Turnover	5.38	4.77	4.57	4.44	5.34
9. Sales per Dollar Working Capital	6.74	6.33	7.02	7.61	−45.77
10. Retained Earning/ Total Assets	0.32	0.30	0.01	−0.01	−0.26
11. Total Assets	0.94	.76	0.39	0.45	0.43



LEGENDA
1. Curvature of mouth
2. Eccentricity of lower ellipse
3. Eccentricity of eyes
4. Half-length of eyes
5. Ear diameter
6. Angle of eyebrow
7. Position of pupils
8. Length of eyebrow
9. Eccentricity of upper ellipse
10. Length of nose
11. Width of nose

Use of Chernoff faces

- Limitations
  - different variables are not easily comparable
  - a legenda is needed
  - nominal variables can not be represented
- It is advisable to couple a tabular view with the faces

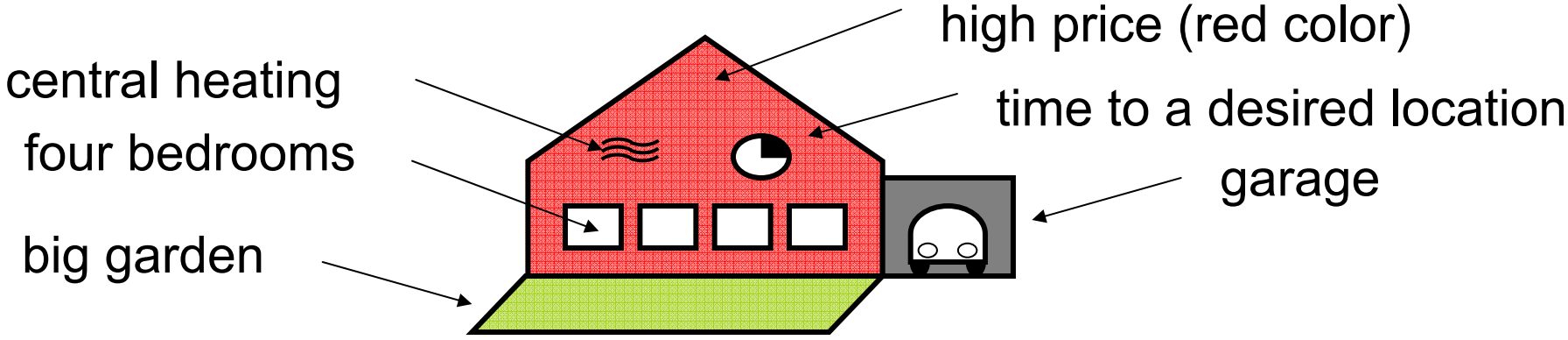


Michael Porath

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Multidimensional icons

- [Spence & Parr 91] proposed multidimensional icons to encode multivariate data
  - the application was the search through dwelling offers
- Eight attribute values are encoded in a glyph
  - color encodes price, shape encodes type, the number of visible windows encodes the number of bedrooms, etc

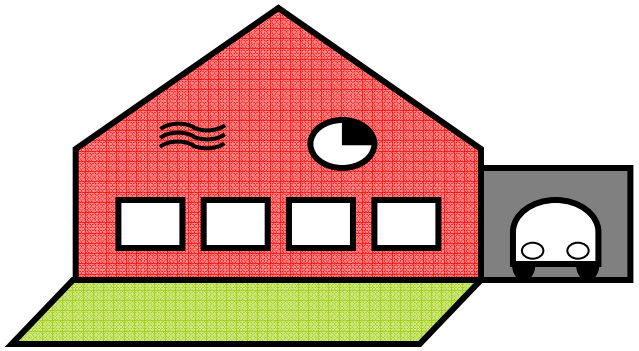


- Allows quick filtering of uninteresting offers

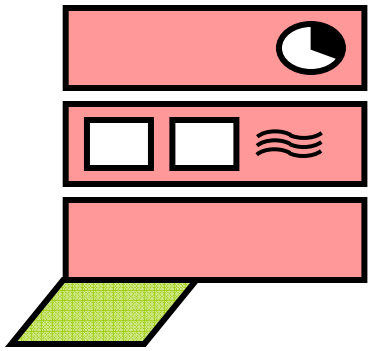
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Multidimensional icons

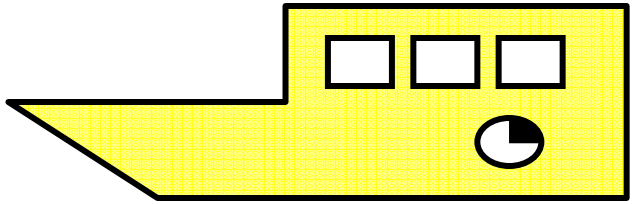
- Examples



house
£400,000
garage
central heating
four bedrooms
good repair
large garden
Victoria 15 mins



flat
£300,000
no garage
central heating
two bedrooms
poor repair
small garden
Victoria 20 mins



houseboat
£200,000
no garage
no central heating
three bedrooms
good repair
no garden
Victoria 15 mins

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Multidimensional icons

- Strategy: encode variables value in petals size
  - Moritz Stefaner

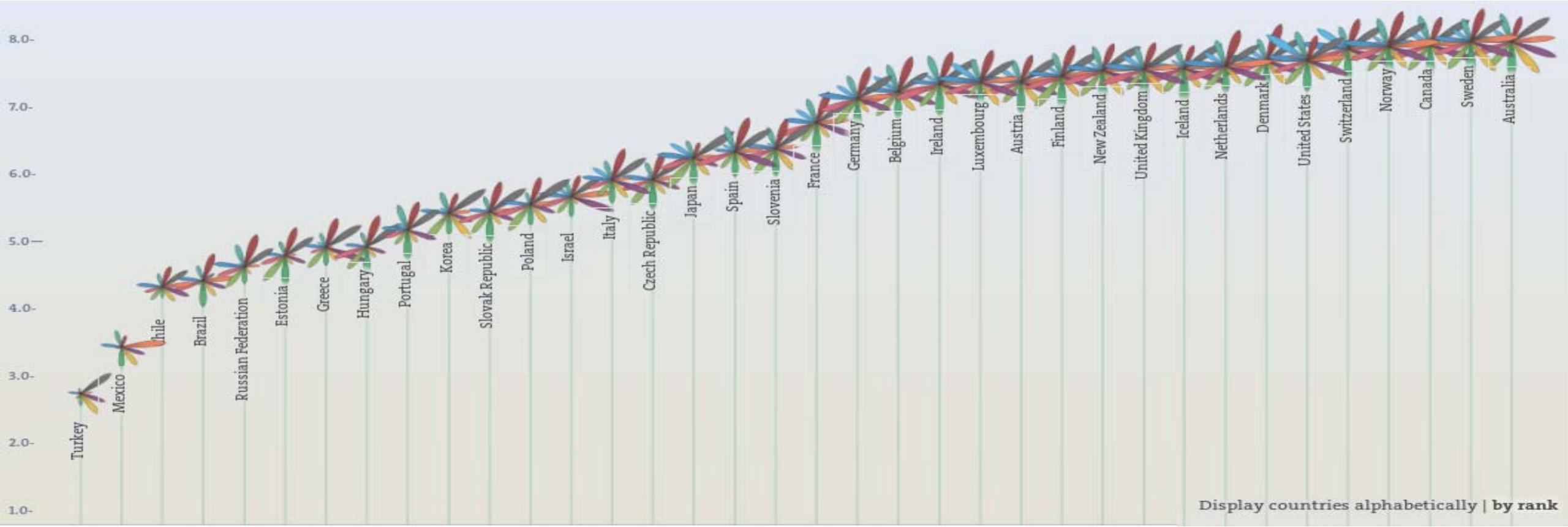




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All countries ordered by rank

- See http://www.oecdbetterlifeindex.org/
  - Moritz Stefaner site at http://moritz.stefaner.eu/



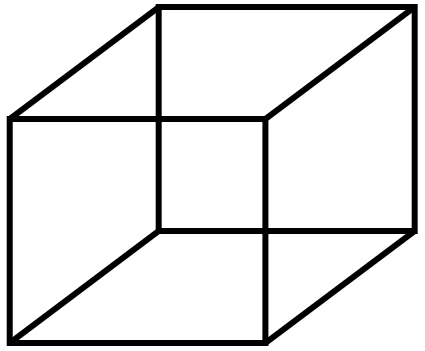
 copyright ©2026 m.patrignani

Parallel coordinates

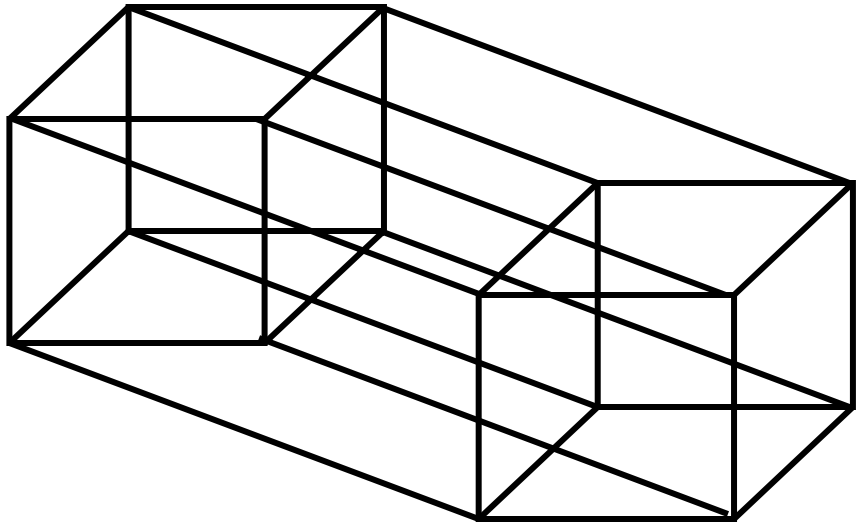
- Limitation of Cartesian coordinates
  - you cannot increase the number of dimensions



2D



3D

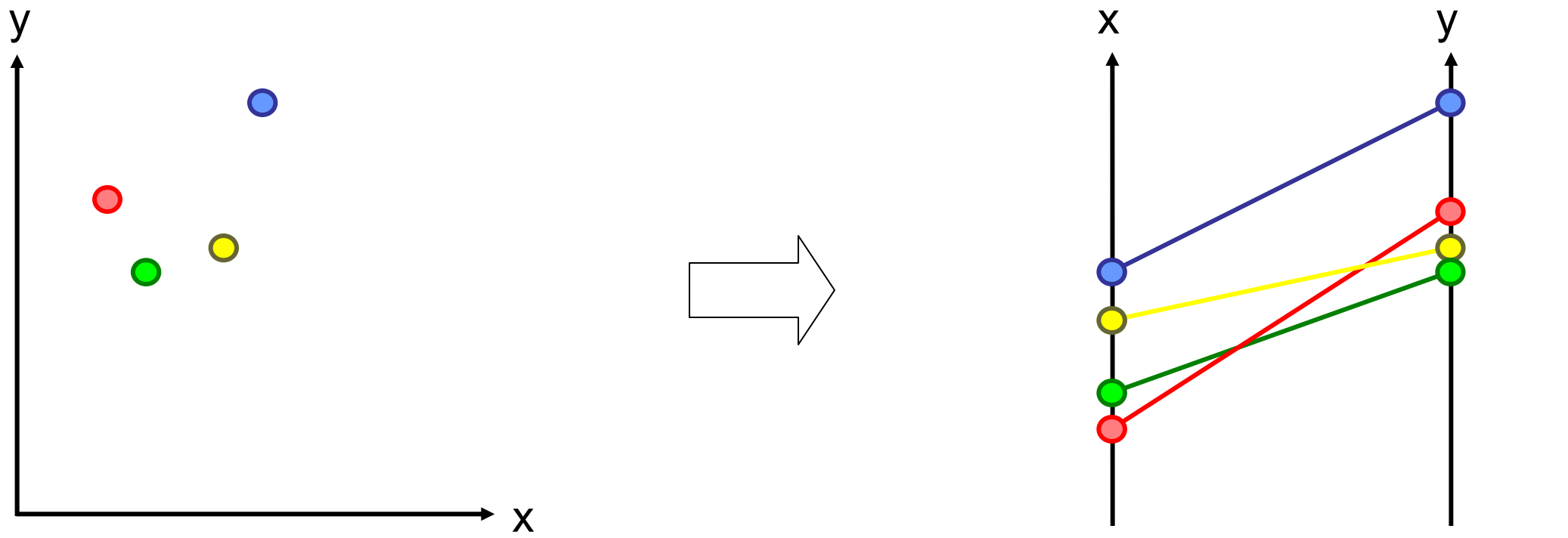


4D

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Parallel coordinates

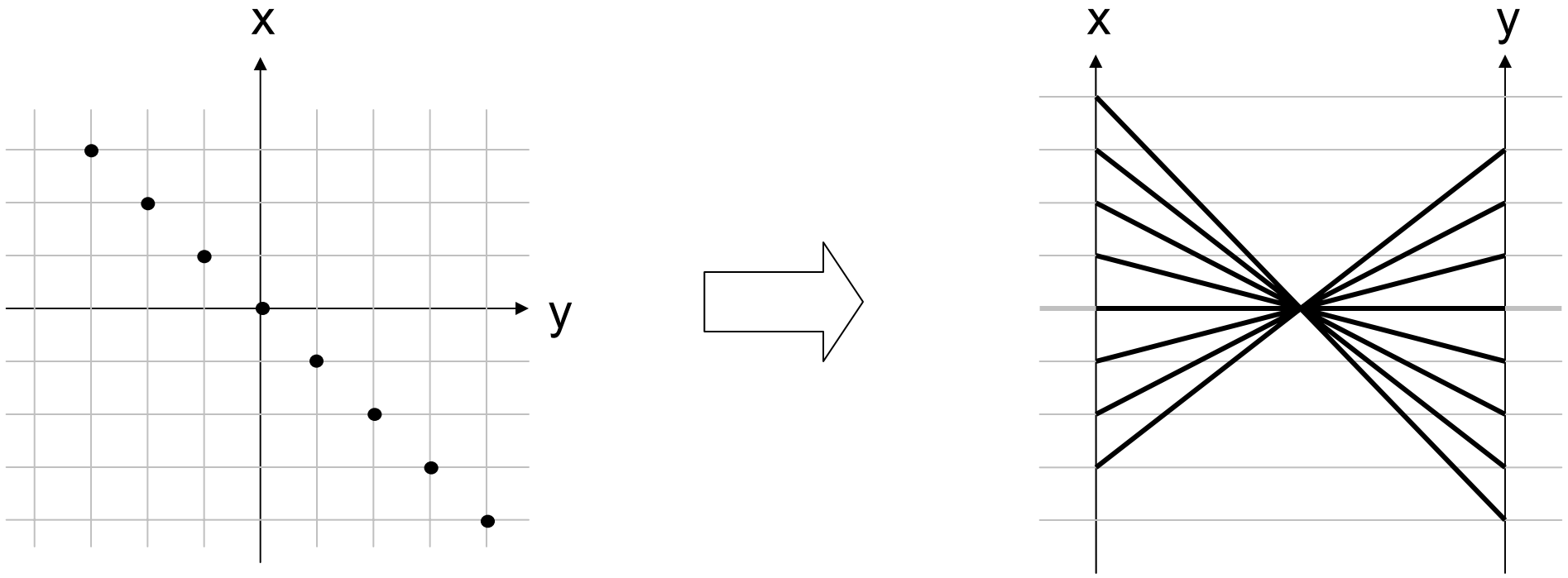
- A point in the Cartesian plane is a line in parallel coordinates



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Parallel coordinates

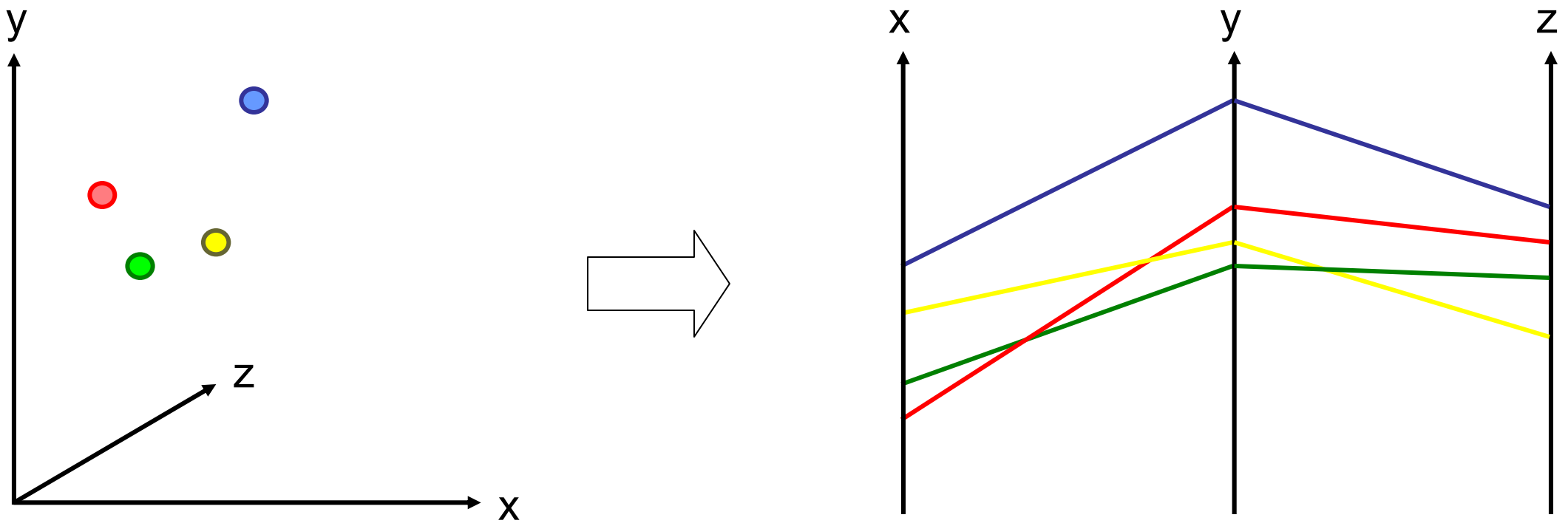
- Points that are aligned in the Cartesian plane correspond to crossing lines in parallel coordinates



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Adding a third axis

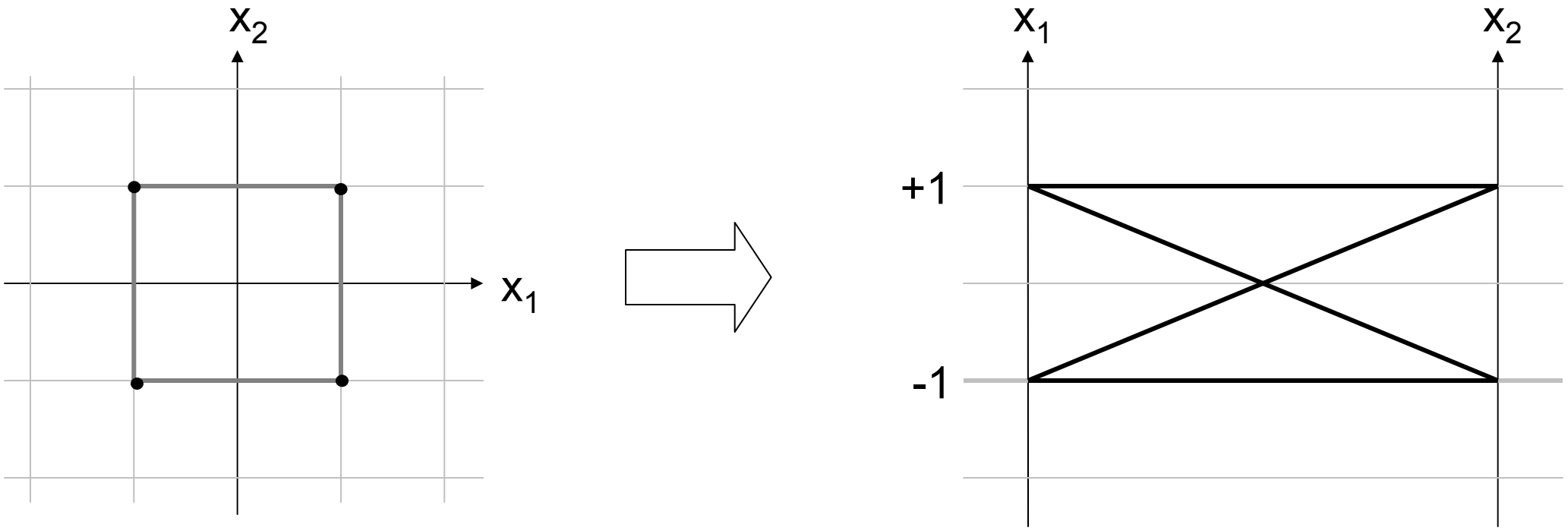
- You can add further axes with the same strategy



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Parallel coordinates

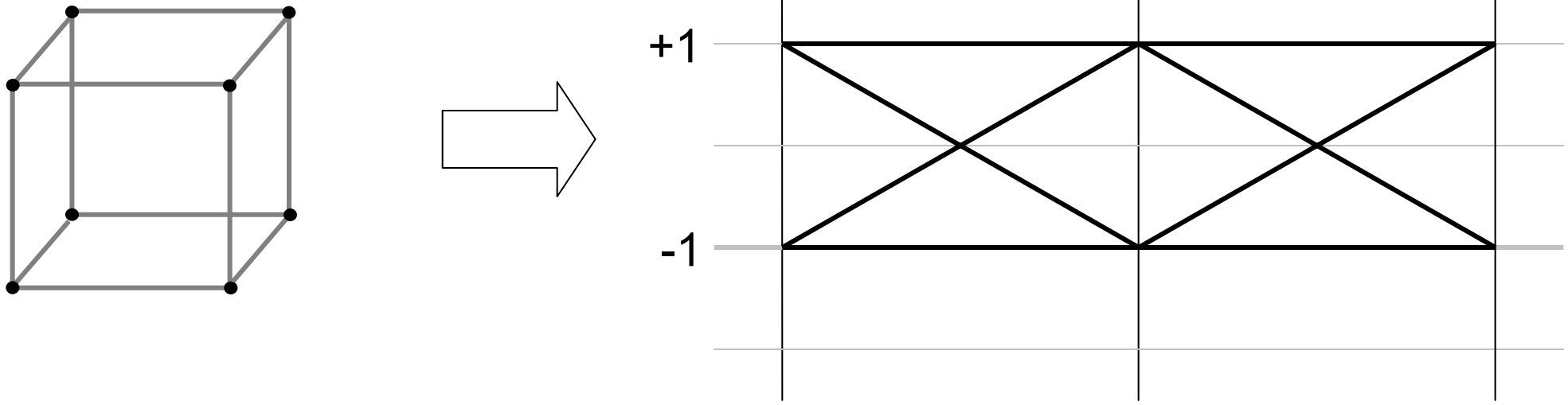
- The four vertices of a square are represented by four lines in parallel coordinates



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Parallel coordinates

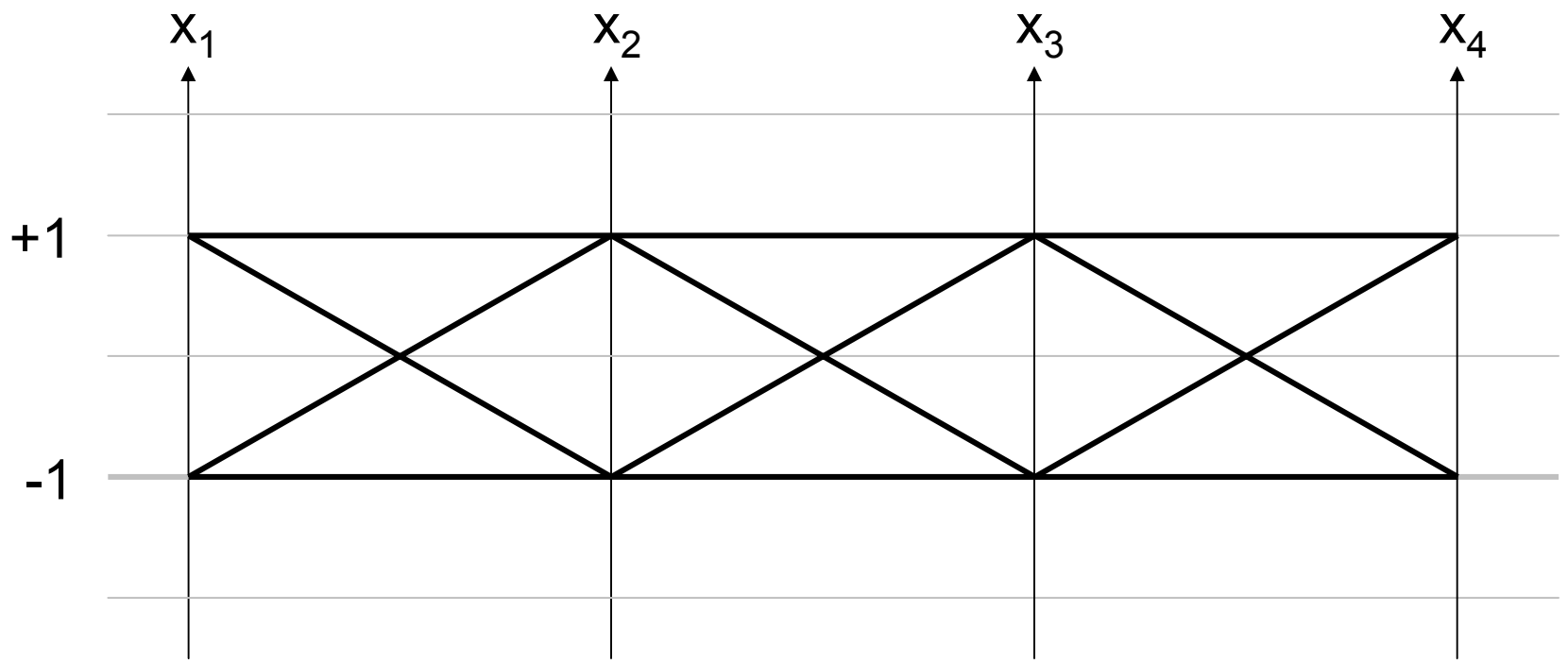
- The eight vertices of a cube in parallel coordinates



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Parallel coordinates

- The sixteen vertices of a hypercube in parallel coordinates

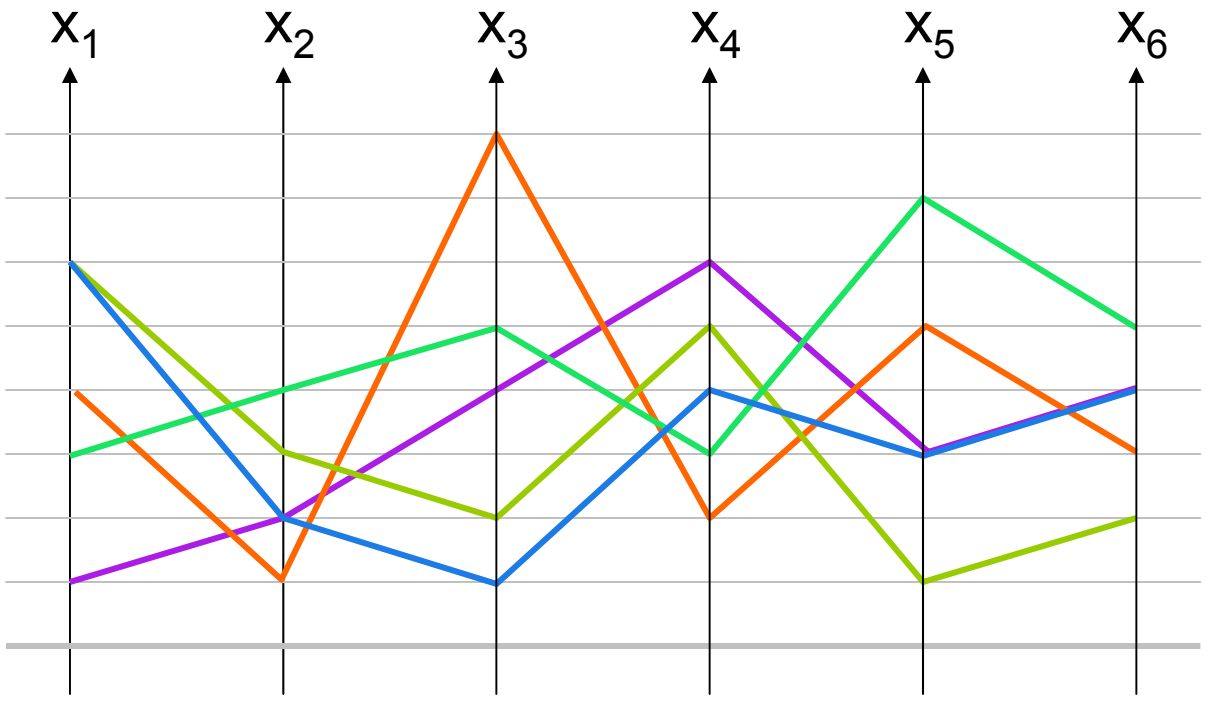


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Parallel coordinates

- Multiple variables can be represented

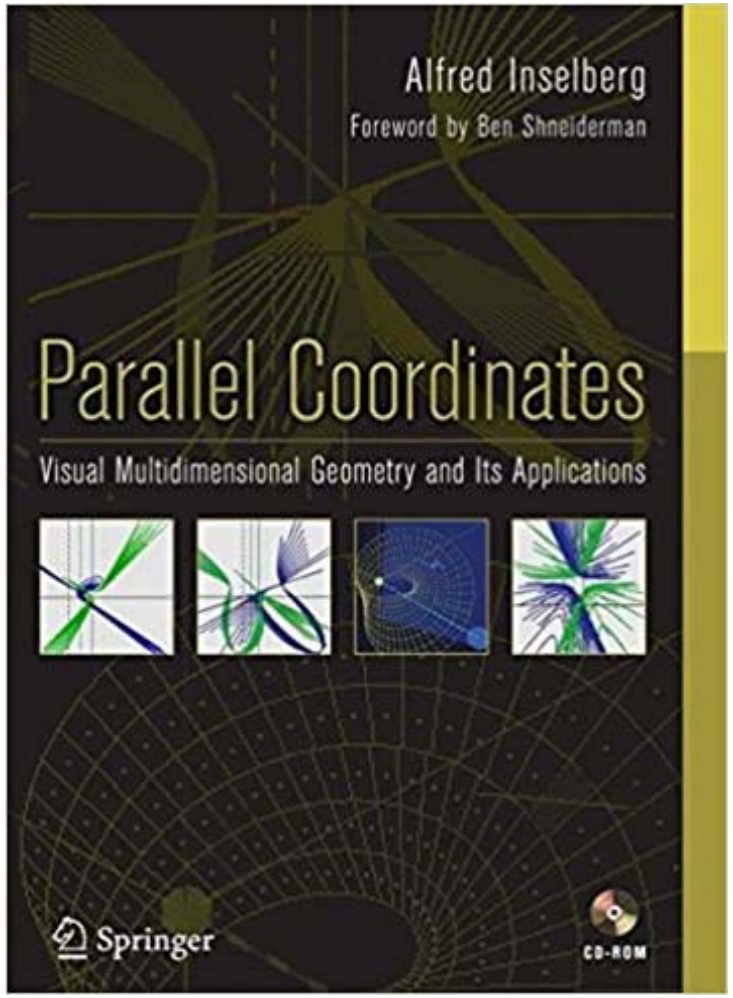
	V_1	V_2	V_3	V_4	V_5	V_6
C_1	4	1	8	3	5	3
C_2	6	3	2	5	1	2
C_3	3	4	5	3	7	5
C_4	6	2	1	4	3	4
C_5	1	2	4	6	3	4



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Parallel coordinates' history



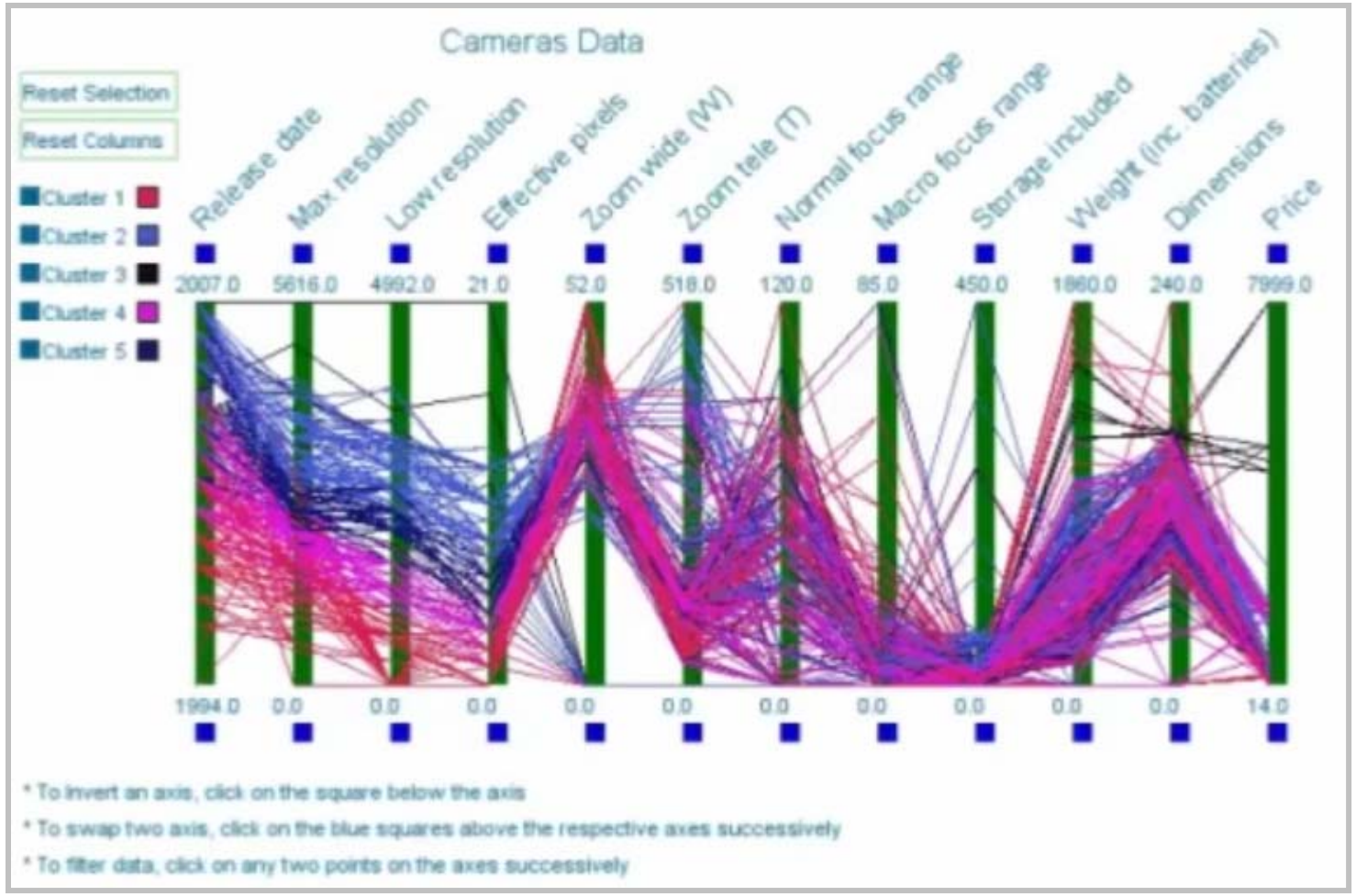


- Due to Alfred Inselberg (1936-2019)
  - "*The Plane with Parallel Coordinates*". Visual Computer 1, 1985

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Example of parallel-coordinates

- See video by Mukund Raj

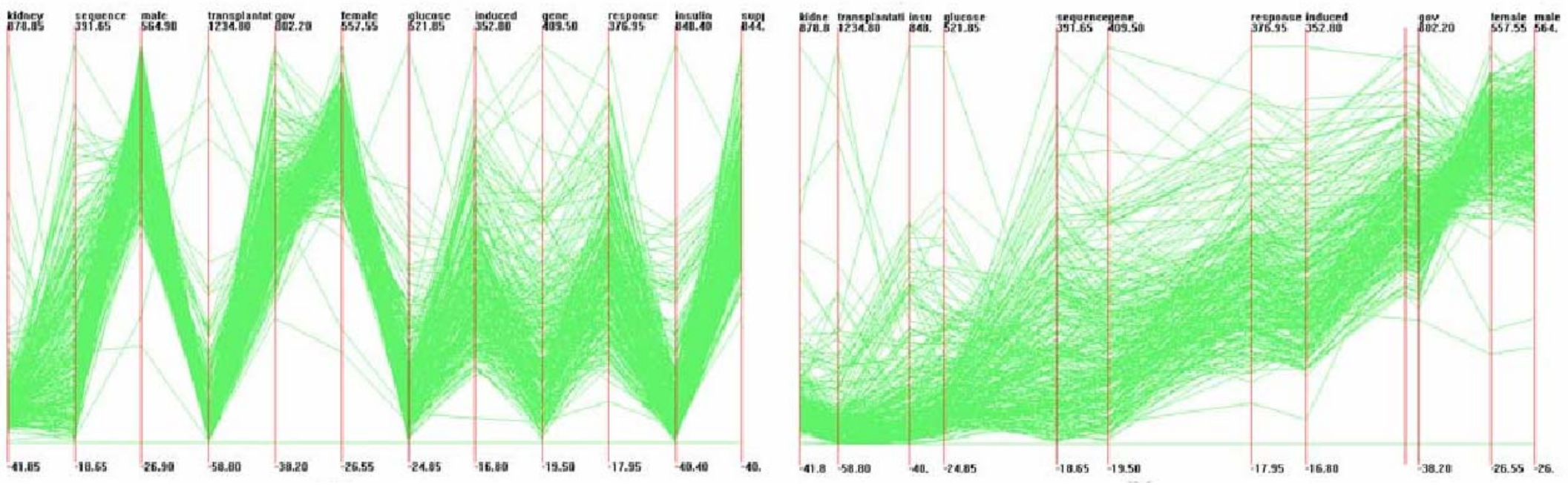


Video at http://www.youtube.com/watch?v=JI1WGzwMEGA

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Reordering dimensions

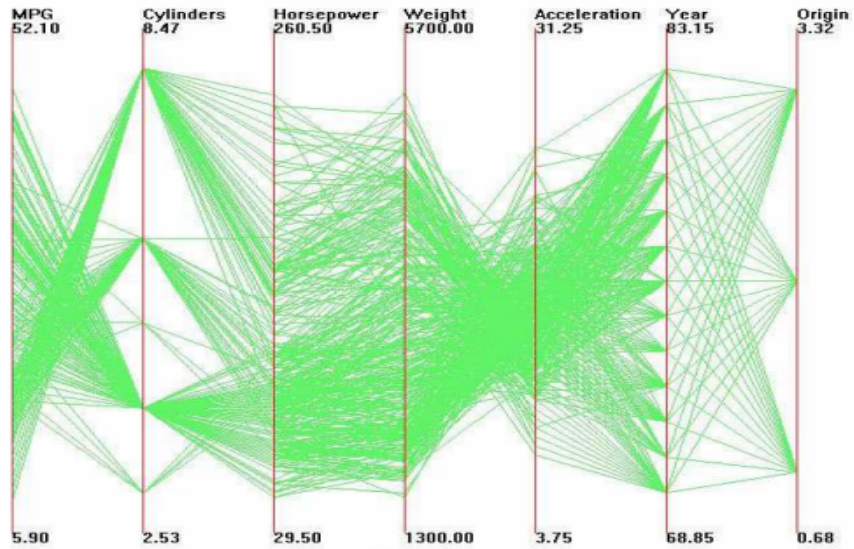
- Automatic reordering (and spacing) of dimensions may reveal hidden trends
  - the same dimension ordered according to similarity
    - from [Yang *et al.* 03]

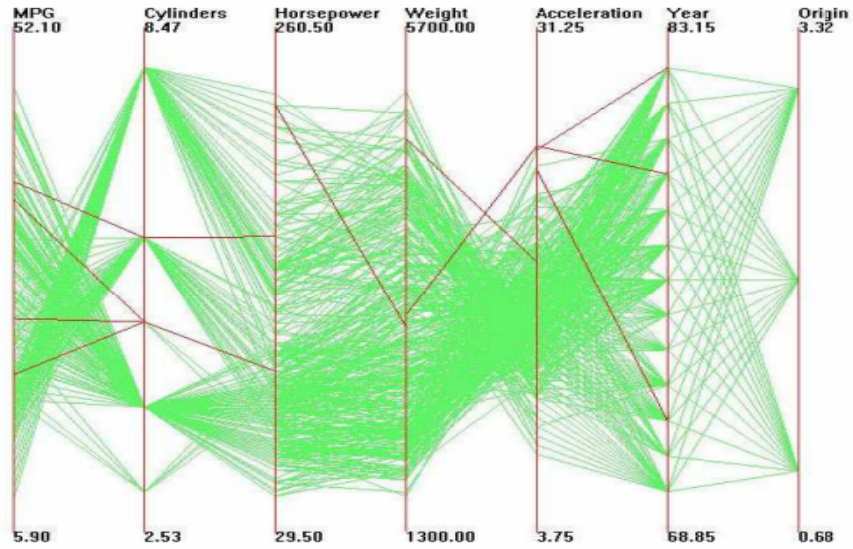


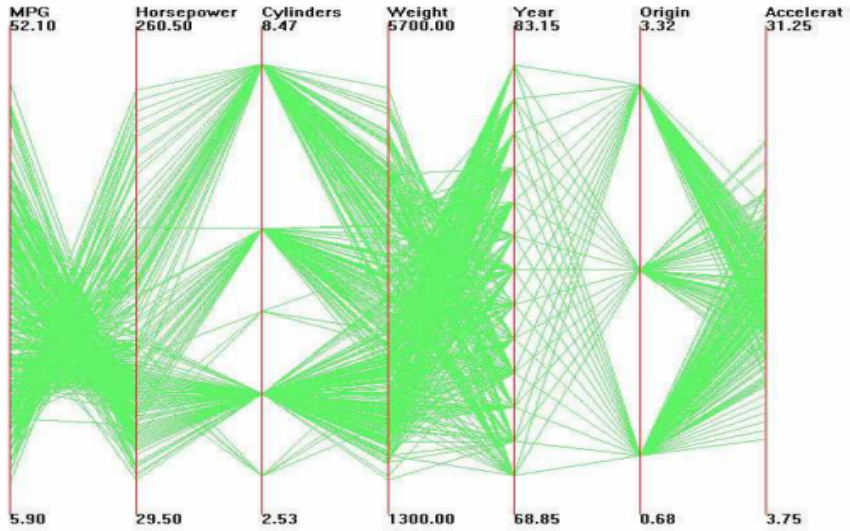
 copyright ©2026 m.patrignani

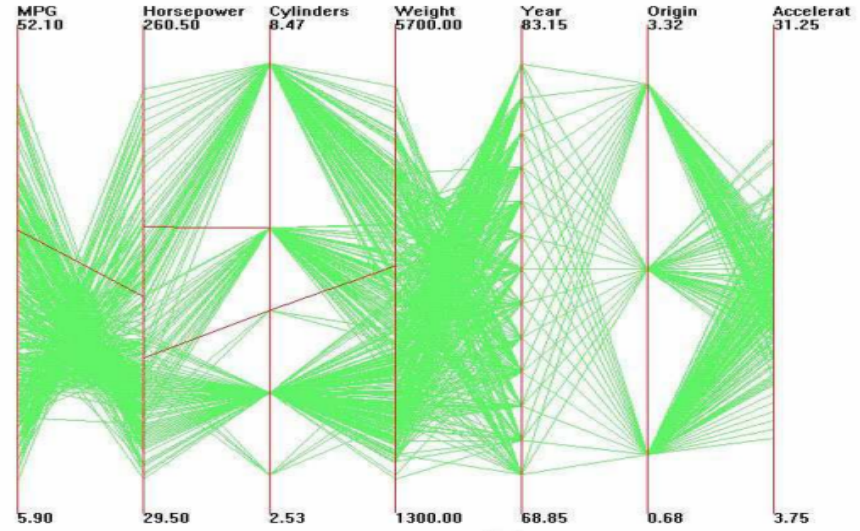
Reordering dimensions to reduce cluttering

- Outliers shown red on the right charts
- The lower charts show reordered dimensions
- Reordering reduces the number of outliers
  - from [Peng *et al.* 04]





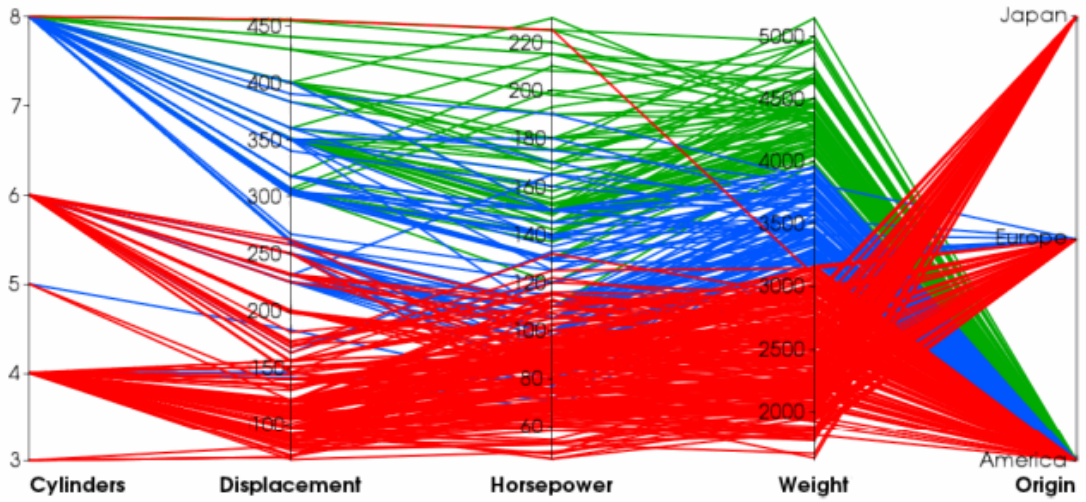


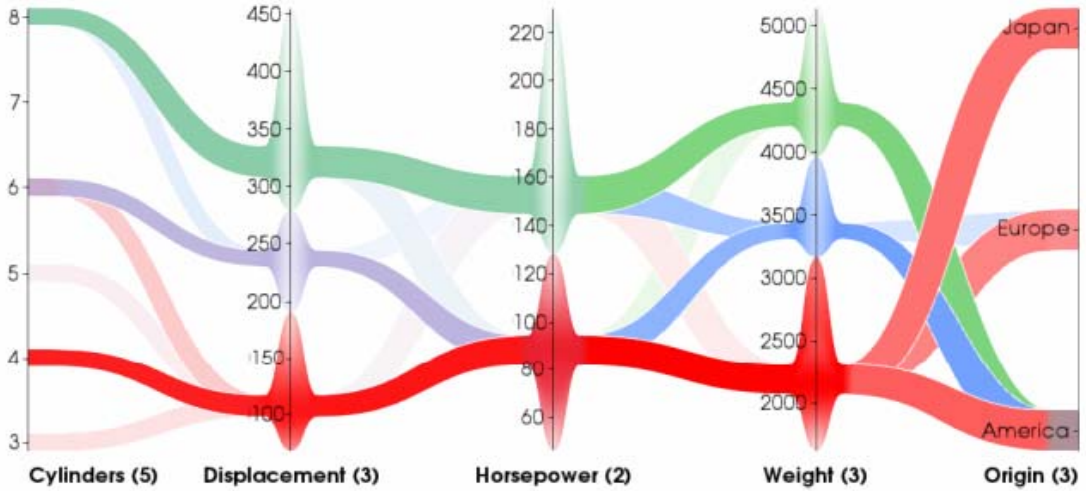


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Edge bundling in parallel-coordinates

- Edge-bundling using density-based clustering in parallel coordinates
  - from [Palmas *et al.* 14]



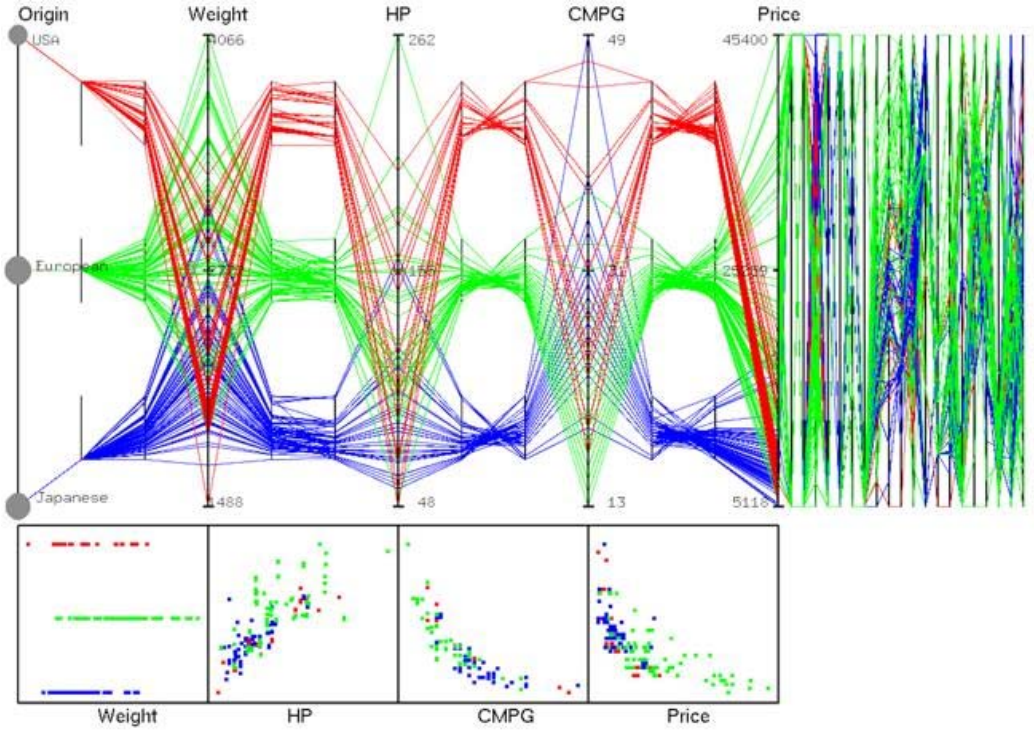


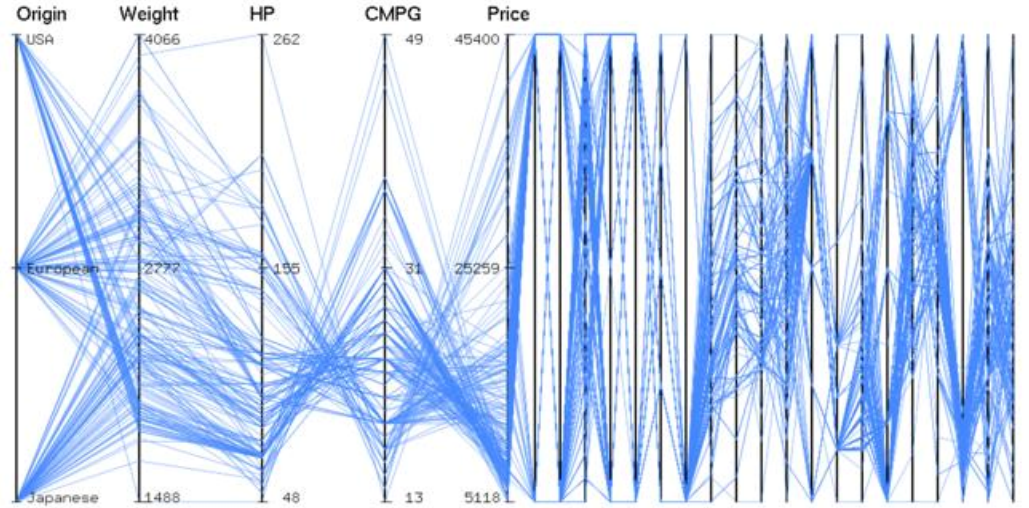
Video at
http://www.youtube.com/watch?v=VK9YbphYg4I

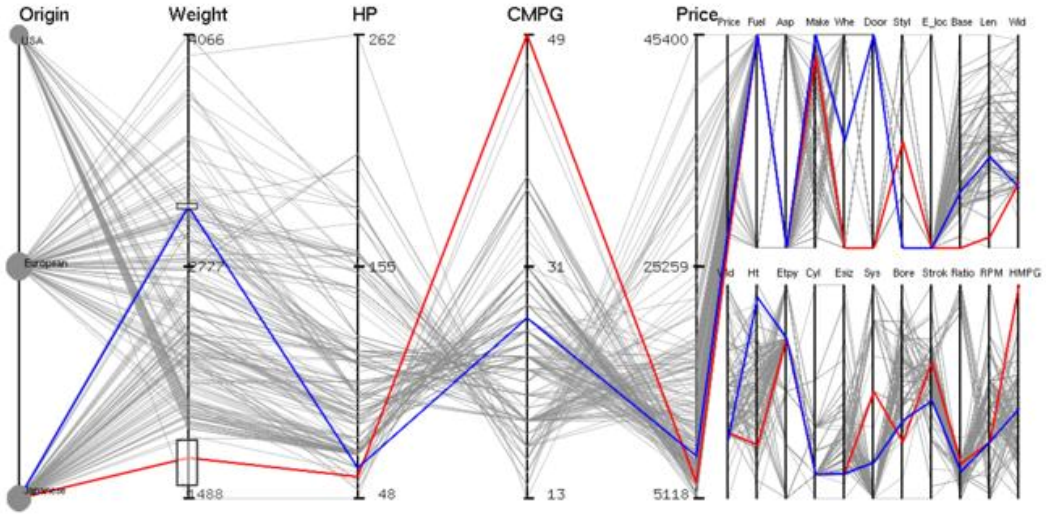
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Bifocal parallel-coordinates

- Proposed by
  [Kaur & Karki, 2018]



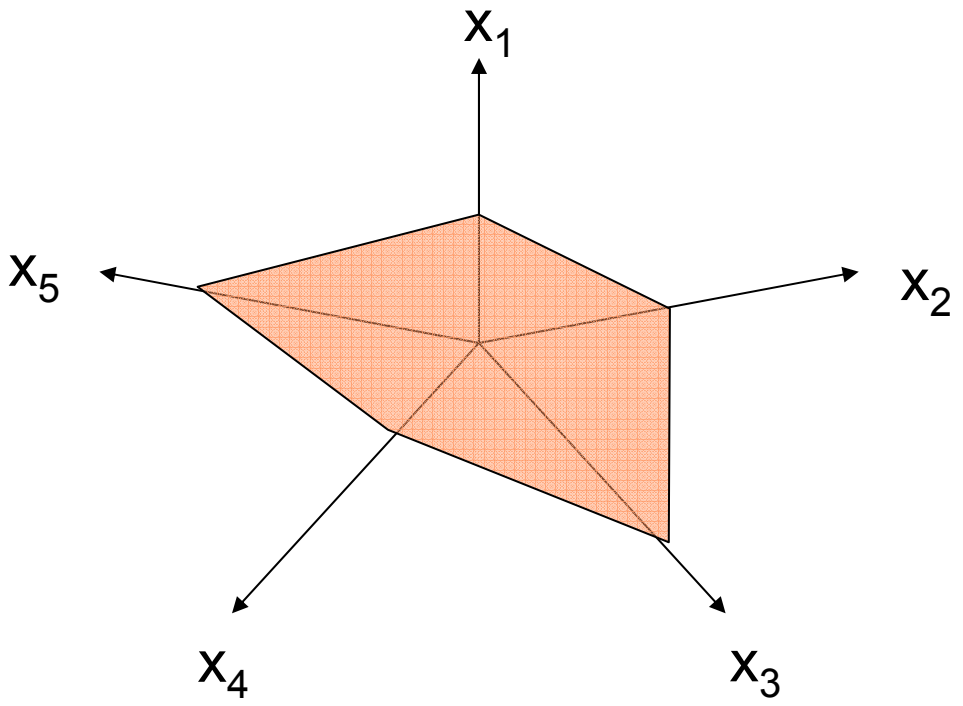




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Star plots

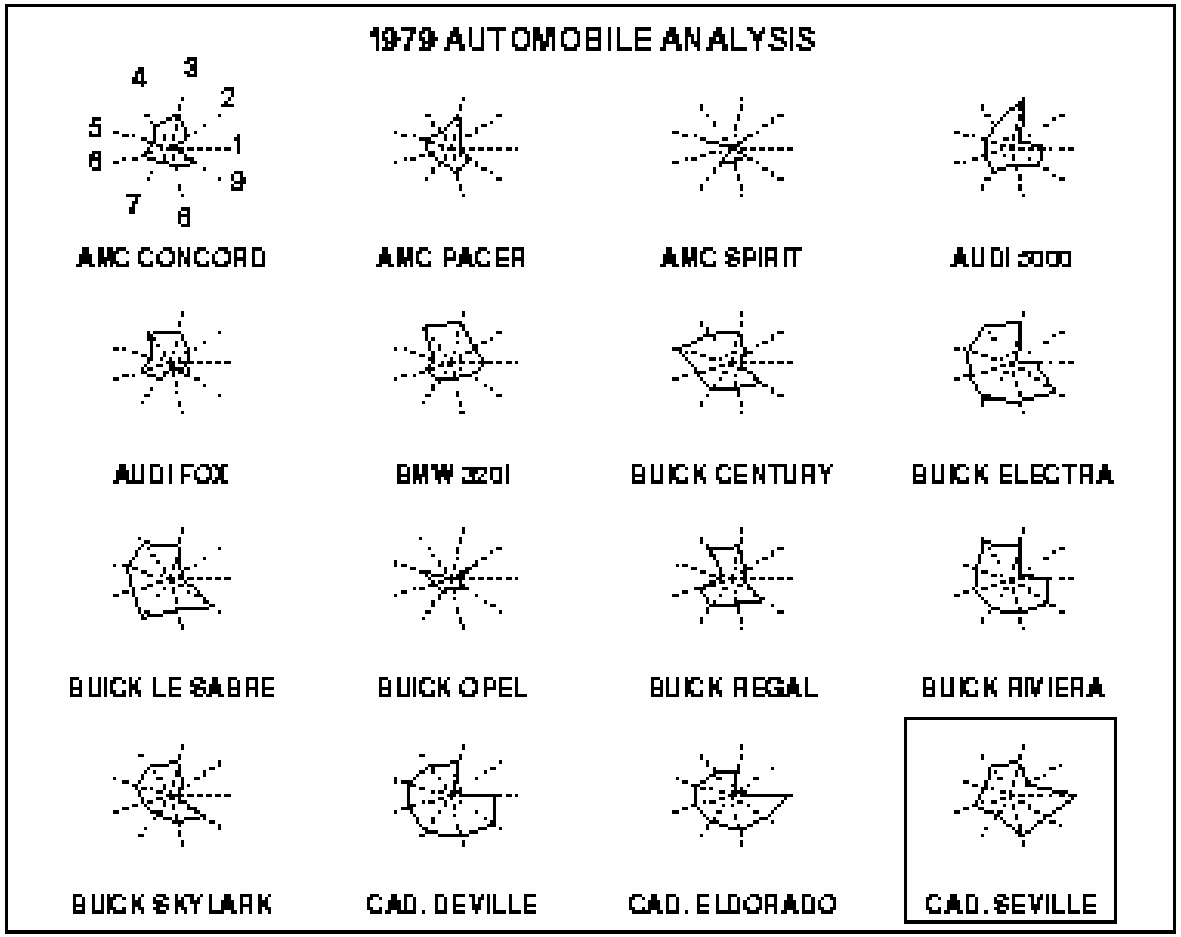
- Also called “star glyphs”, “web charts”, “spider charts”, “radar charts”, or “Kiviat diagrams”
  - [Kolence & Kiviat, 1973]
- Axes cross in the origin and are spaced around a cycle
  - each spoke encodes a variable
  - each data case is a shape



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Star plot example

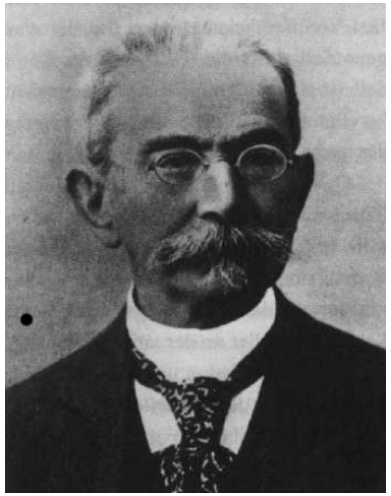
- This star plot represents the characteristics of 15 cars
- The variable list for the sample star plot is
  1. Price
  2. Mileage (MPG)
  3. 1978 Repair Record
     - 1 = Worst, 5 = Best
  4. 1977 Repair Record
  5. Headroom
  6. Rear Seat Room
  7. Trunk Space
  8. Weight
  9. Length



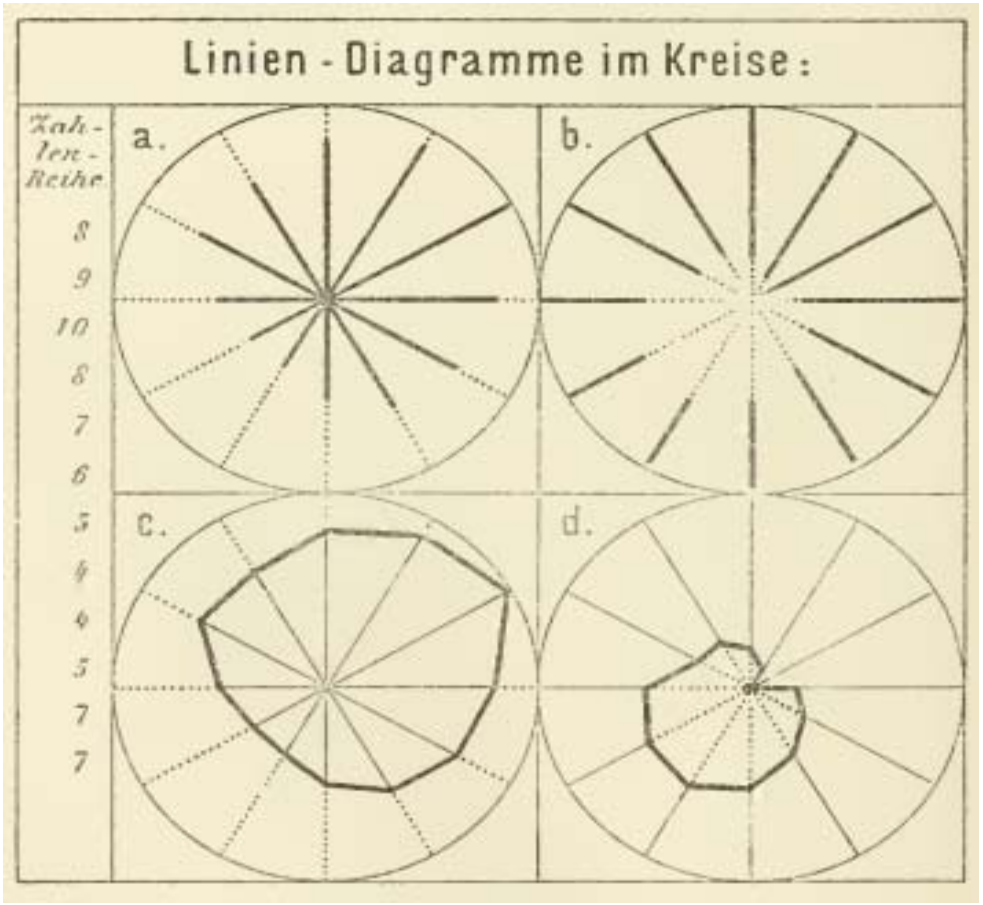
From http://www.itl.nist.gov/div898/handbook/eda/section3/starplot.htm

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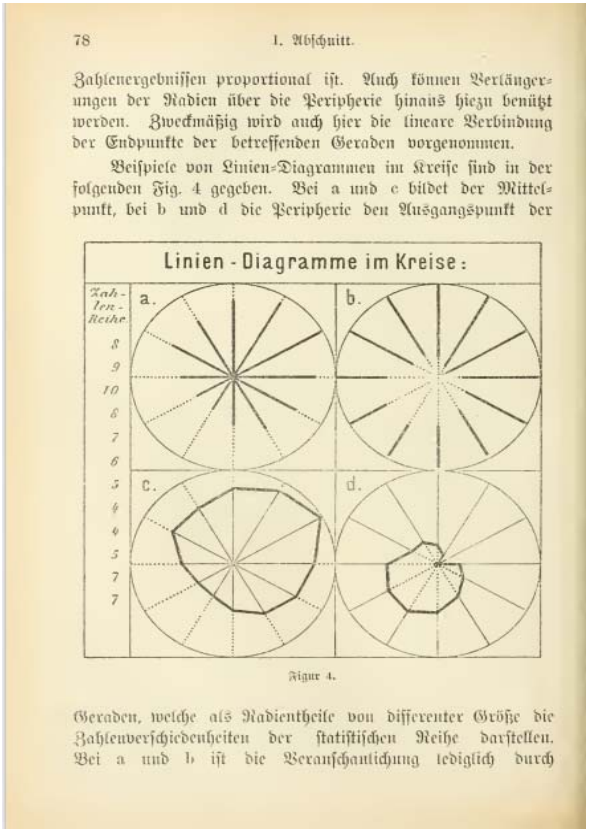
Star plots history



- Introduced by Georg von Mayr in 1877



page 78 of “Die Gesetzmässigkeit im Gesellschaftsleben statistische Studien“



78 I. Abschnitt.

Zahlenergebnissen proportional ist. Auch können Verlängerungen der Radien über die Peripherie hinaus hiezu benützt werden. Zweckmäßig wird auch hier die lineare Verbindung der Endpunkte der betreffenden Geraden vorgenommen.

Beispiele von Linien-Diagrammen im Kreise sind in der folgenden Fig. 4 gegeben. Bei a und c bildet der Mittelpunkt, bei b und d die Peripherie den Ausgangspunkt der

Linien - Diagramme im Kreise:

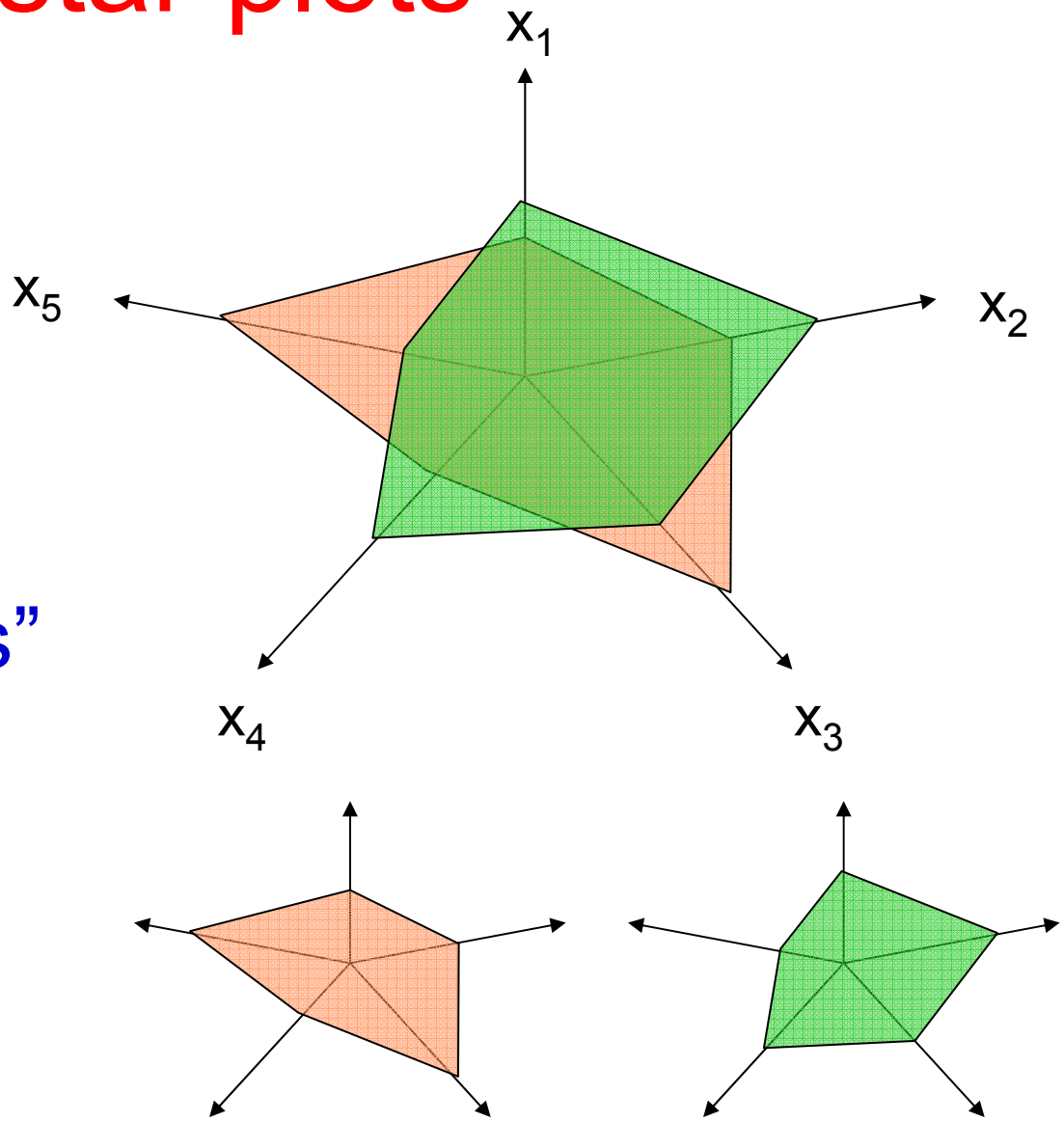
Figur 4.

Geraden, welche als Radientheile von differenter Größe die Zahlenverschiedenheiten der statistischen Reihe darstellen. Bei a und b ist die Veranschaulichung lediglich durch

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Comparing star plots

- Usually viewed with overlapped shapes or in small multiples
- The order of the axes is not significant
- If some axis is omitted they take the name of “glyph plots” or “star glyph”
- Advantages
  - good at comparing data cases when variables have an inherent priority

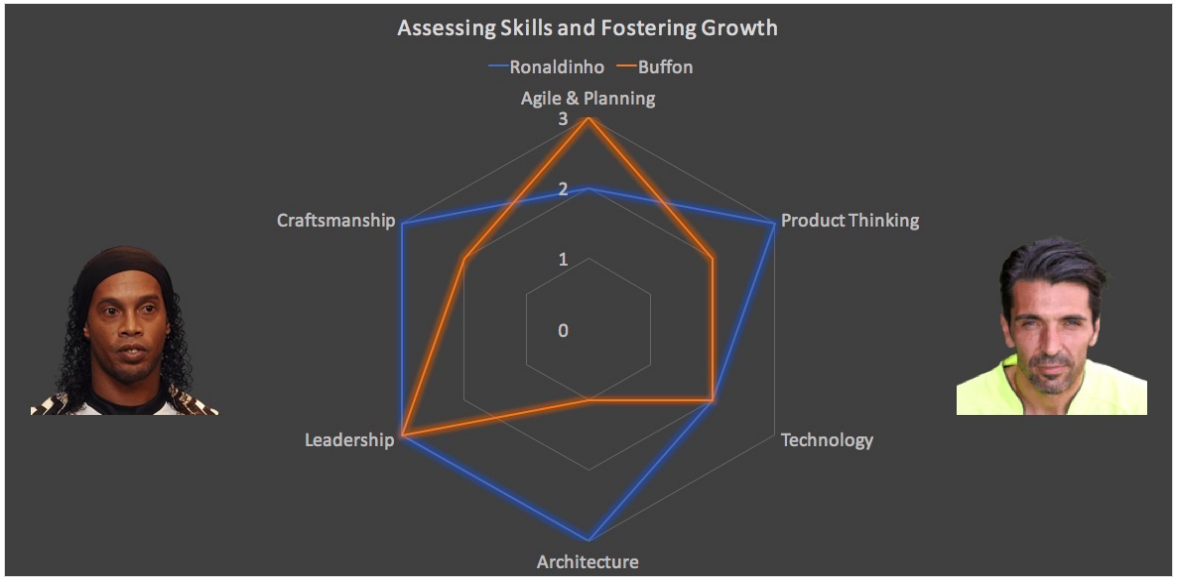


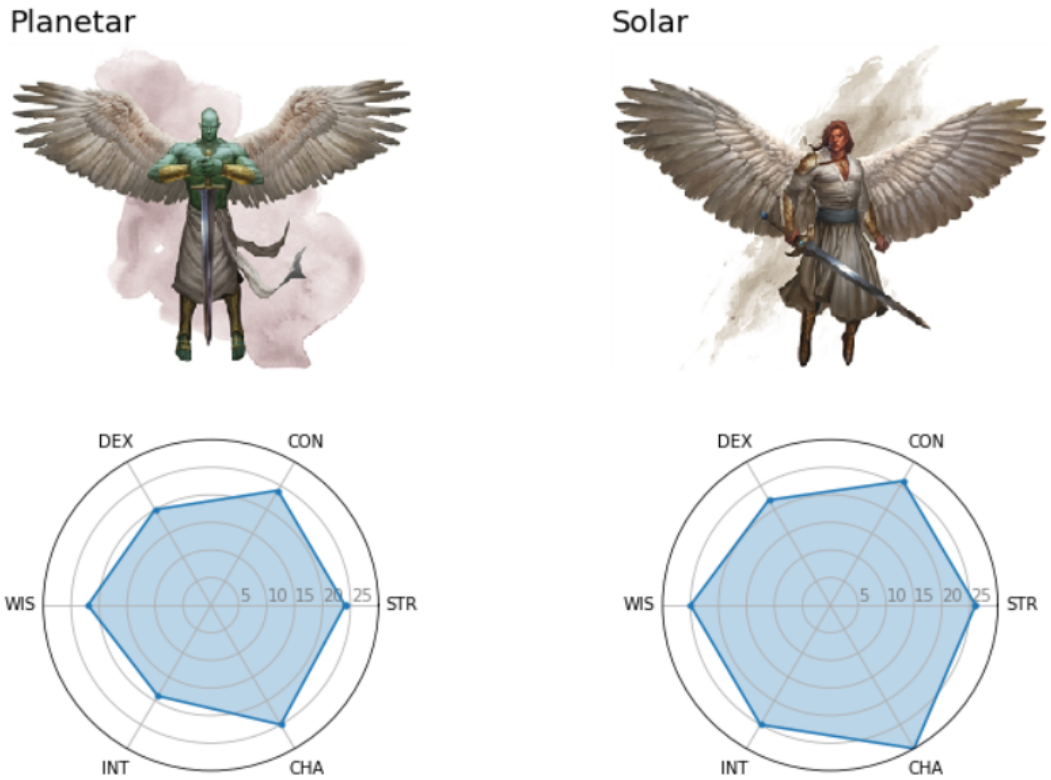
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Examples of star plots comparison

- Sometimes used in videogames

FIFA 17 virtual online sport game



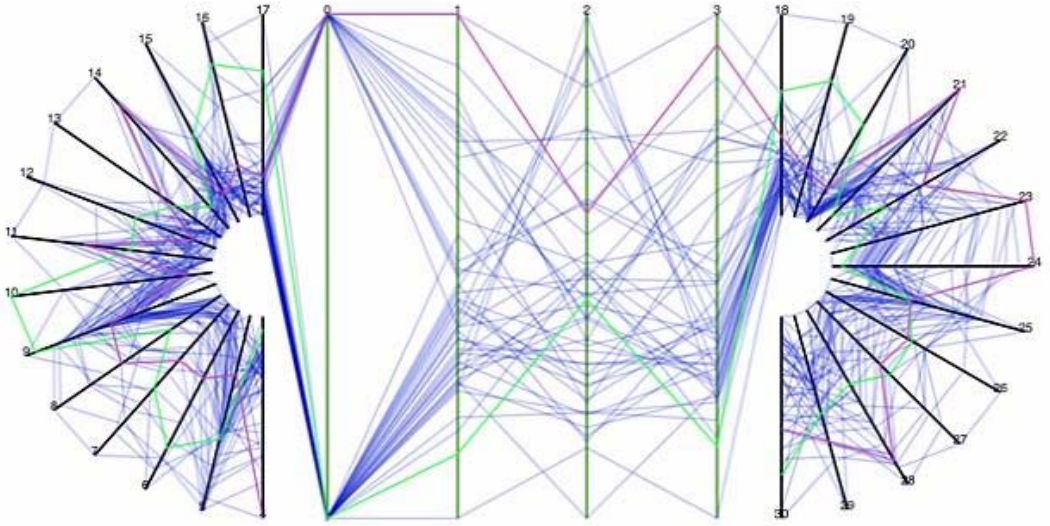


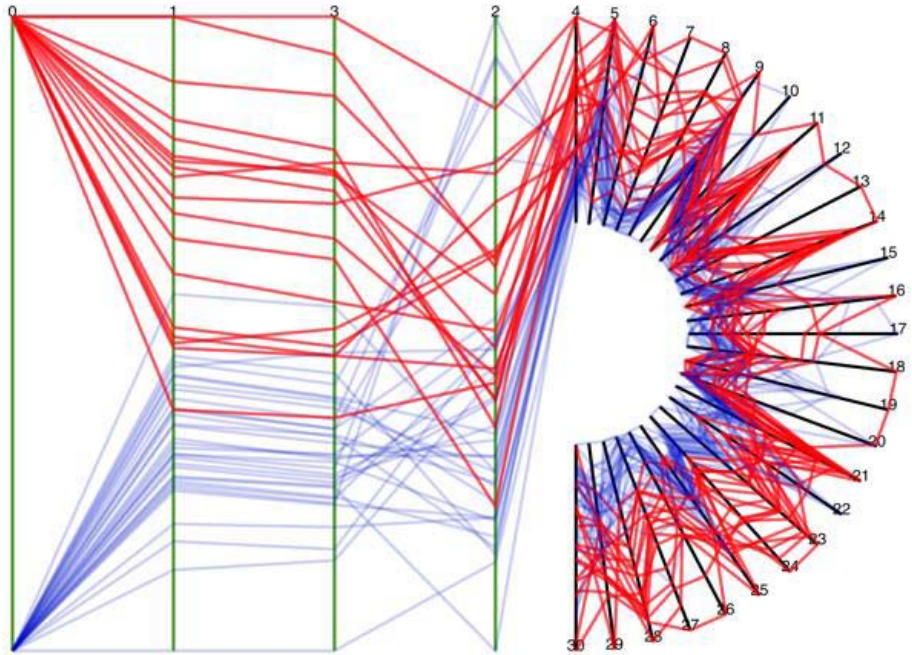
From [Carvalho 2020]

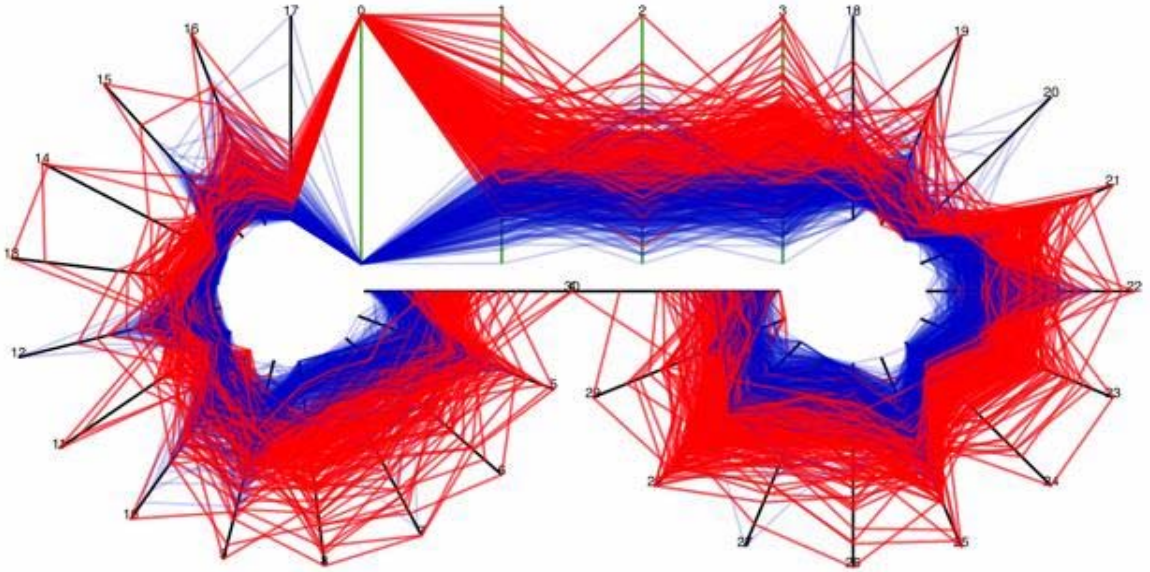
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Combining parallel-coordinates and star plots

- Parallel-coordinates and star plots are actually very similar
  - [Xie & Karki, 2019]



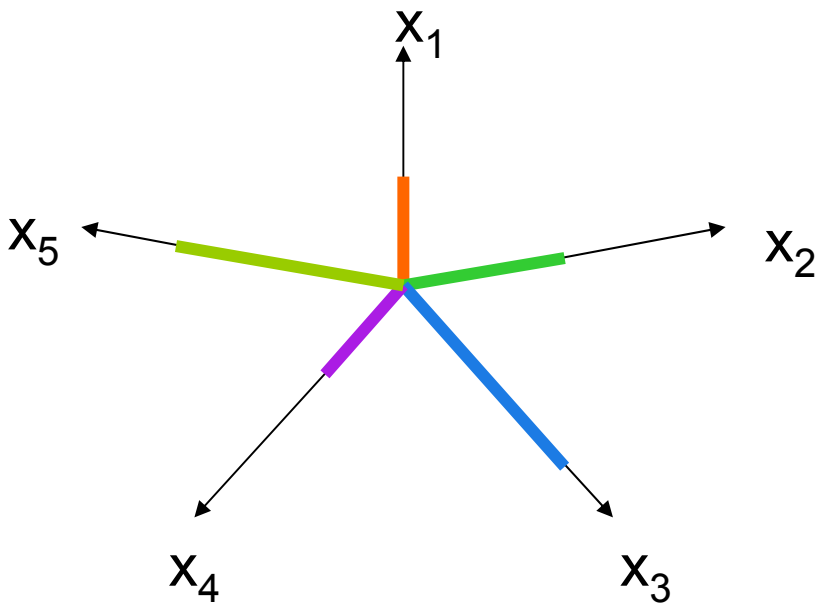


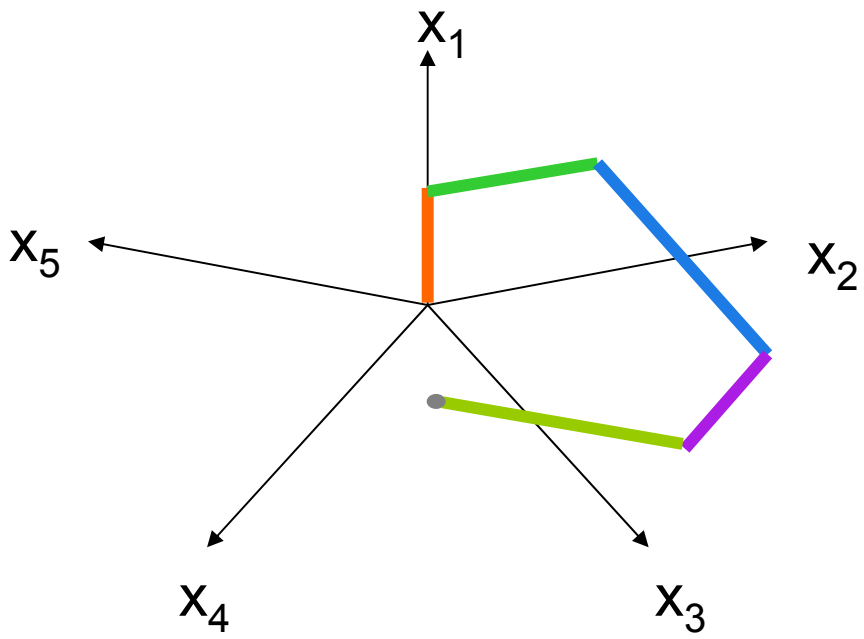


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Star coordinates

- Similar to star plots: accumulate the values by summing up the vectors representing them
  - each spoke encodes a variable
  - each data case is a point

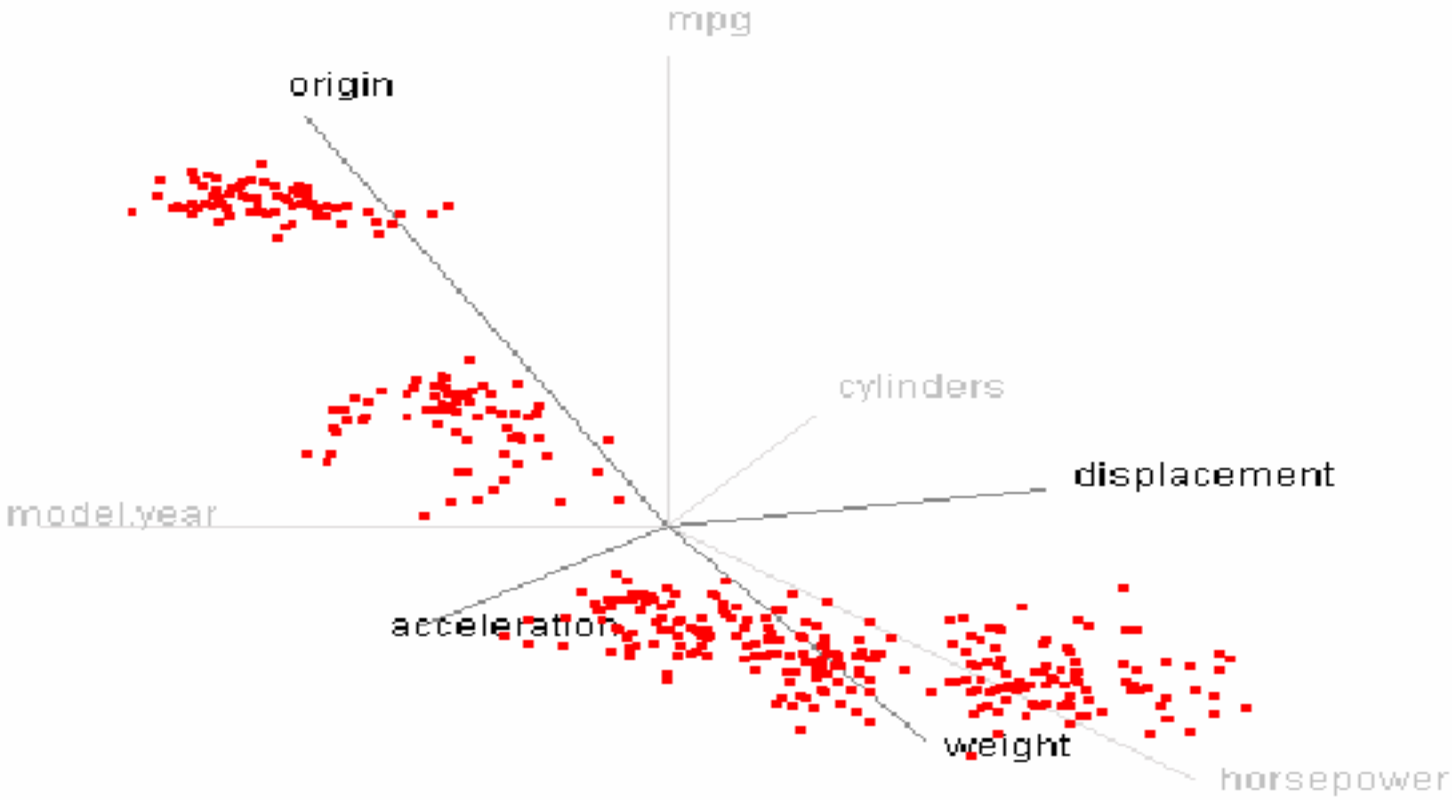




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Star coordinates

- Data cases with similar values will lead to clusters of points
  - but different values may sum up to the same point!
  - introduced by [Kandogan 00]



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Bibliography and further readings

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- [Chernoff 73] Herman Chernoff, "The Use of Faces to Represent Points in K-Dimensional Space Graphically“, 1973
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